

FUNDAÇÃO GETULIO VARGAS
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IGOR PATRÍCIO MICHELS

BRAZILIAN FOOTBALL MODELLING: CALIBRATED BAYESIAN
MODELS AT CLUB AND PLAYER LEVELS

Rio de Janeiro
2026

IGOR PATRÍCIO MICHELS

**BRAZILIAN FOOTBALL MODELLING: CALIBRATED BAYESIAN
MODELS AT CLUB AND PLAYER LEVELS**

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of Applied Mathematics (FGV/EMAp)
to obtain the Master's degree in Applied
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Adviser: Luiz Max F. de Carvalho

Rio de Janeiro

2026

I dedicate this dissertation to my grandmother Ângela
(*in memoriam*).

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Bazinga!

Sheldon Cooper (The Big Bang Theory)

Growth occurs when one surpasses their limits.

Rock Lee (Naruto)

Abstract

Probabilistic models for football outcomes have been extensively studied for European competitions, but applications to Brazilian football remain limited. This dissertation introduces a comprehensive dataset of Brazilian football matches spanning 2013 to 2025, covering the four national divisions and the *Copa do Brasil*, with detailed information on players, goals, cards, and substitutions. Based on this dataset, we develop a collection of Bayesian models from the Bradley-Terry and Poisson families, formulated at both club and player levels. All models are validated through Simulation-Based Calibration to ensure that the inference procedure correctly approximates the posterior distribution. The validated models are then applied to *Brasileirão Série A* (the Brazilian first division) data, where their predictive performance is compared using proper scoring rules. Finally, we extend the best-performing club-level specifications to the player level, enabling a more granular analysis of team performance. All data and code are publicly available to support reproducibility.

Keywords: Brazilian football, Bayesian inference, Bradley-Terry model, Poisson model, Simulation-Based Calibration, player-level modelling.

Resumo

Modelos probabilísticos para resultados de futebol têm sido amplamente estudados para competições europeias, porém aplicações ao futebol brasileiro permanecem limitadas. Este trabalho apresenta um conjunto de dados abrangente de partidas do futebol brasileiro entre 2013 e 2025, cobrindo as quatro divisões nacionais e a Copa do Brasil, com informações detalhadas sobre jogadores, gols, cartões e substituições. Com base nesse conjunto de dados, desenvolvemos uma coleção de modelos bayesianos das famílias Bradley-Terry e Poisson, formulados tanto em nível de clube quanto de jogador. Todos os modelos são validados por meio de Calibração Baseada em Simulação para garantir que o procedimento de inferência reproduza corretamente a distribuição a posteriori. Os modelos validados são então aplicados aos dados do Brasileirão Série A (a primeira divisão do futebol brasileiro), onde seu desempenho preditivo é comparado utilizando regras de pontuação próprias. Por fim, estendemos as especificações de nível de clube com melhor desempenho para o nível de jogador, permitindo uma análise mais granular do desempenho das equipes. Todos os dados e códigos estão disponíveis publicamente para garantir a reprodutibilidade.

Palavras-chave: futebol brasileiro, inferência bayesiana, modelo Bradley-Terry, modelo Poisson, Calibração Baseada em Simulação, modelagem em nível de jogador.

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1 Background

Football is the most popular sport in the world. Originating in England during the nineteenth century, the game rapidly spread across continents and became deeply embedded in national cultures. In Brazil, where the sport arrived in 1894, football transcended its role as mere entertainment to become a defining element of national identity. It is a passion that unites millions of people regardless of social background, region, or creed.

This popularity has also made football a significant economic force. Beyond match-day revenues, merchandise sales, and broadcasting rights, the global sports betting market has grown substantially in recent years. Football accounts for the largest share of this market, which is projected to reach approximately 187 billion dollars by 2030 ([GRAND VIEW RESEARCH, 2024](#)). Questions such as “which team is better?” or “is player A superior to player B?” are no longer confined to casual conversations among fans. They now carry financial implications for betting operators and bettors alike, creating demand for rigorous quantitative analysis.

This context has motivated the development of probabilistic models capable of estimating team strengths, predicting match outcomes, and quantifying the uncertainty associated with these predictions. Such models provide a framework for transforming historical match data into actionable insights, whether for sports analysts, club managers, or betting markets. This chapter reviews the main modelling approaches proposed in the literature, covering paired comparison models, goal-based formulations, player-level extensions, and a simulation-based validation technique.

Among paired comparison models, the Bradley-Terry model ([BRADLEY; TERRY, 1952](#)) was originally developed for analysing experiments where items are compared in pairs and a preference or outcome is recorded. In the context of sports, this model provides a natural framework for modelling match outcomes by relating the probability of one team beating another to latent skill parameters. The original formulation considered binary outcomes, representing a win for one of the two items being compared. [Davidson \(1970\)](#) extended the Bradley-Terry model, introducing a tie parameter. This parameter allows the model to assign non-zero probability to draw outcomes, making it more suitable for football applications. [Davidson and Beaver \(1977\)](#) further extended the model to incorporate order effects, which can capture home effects in sports contexts. These extensions transformed the Bradley-Terry model from a simple binary comparison tool into a flexible framework for modelling sports competitions. More recently, [Cattelan, Varin, and Firth \(2013\)](#) developed dynamic Bradley-Terry models that allow team strengths to evolve over time, capturing the temporal dynamics of team performance throughout a season. This work demonstrated the flexibility of the Bradley-Terry framework for modelling sports tournaments and showed

how temporal variation can be incorporated while maintaining the model’s interpretability.

Rather than modelling only match outcomes, an alternative and complementary approach uses the number of goals scored by each team as variables to be modelled. This approach was formalised by [Maher \(1982\)](#), where these variables are modelled as Poisson with rates that depend on team offensive and defensive abilities. We call these Poisson-based formulations as Poisson models, where the focus is modelling explicitly the final score of a match. [Lee \(1997\)](#) applied Poisson models to English Premier League data, with a particular emphasis on incorporating home effects as a distinct parameter in the model. Independently, [Dixon and Coles \(1997\)](#) extended this class of models by introducing a correlation parameter to account for the dependence between home and away goal counts. Their model incorporated a bivariate Poisson distribution to capture this correlation, building upon earlier work on bivariate Poisson distributions ([HOLGATE, 1964](#)).

[Karlis and Ntzoufras \(2003\)](#) further developed the bivariate Poisson approach, demonstrating that accounting for correlation improves model fit and predictive performance. [Karlis and Ntzoufras \(2009\)](#) proposed using the Skellam distribution to model goal differences directly, offering an alternative to bivariate Poisson models that focuses on the difference between goal counts rather than the counts themselves. This approach can be particularly useful when the research question focuses on match outcomes rather than exact scorelines. Within a Bayesian framework, [Baio and Blangiardo \(2010\)](#) proposed a hierarchical model for estimating team strengths and predicting match scores, introducing a mixture formulation to address the overshrinkage produced by standard hierarchical priors. Similarly to the dynamic extensions in the Bradley-Terry framework, [Rue and Salvesen \(2000\)](#) introduced a Bayesian Poisson model with time-varying team strengths, and [Koopman and Lit \(2015\)](#) combined the bivariate Poisson formulation with state-space methods to model the evolution of team abilities across a season. Recent applications include Poisson-based analyses of the English Premier League ([NGUYEN, 2021](#)) and double Poisson formulations for international tournaments such as Euro 2020 ([PENN; DONNELLY, 2022](#)).

Most of the literature on football modelling operates at the club level, treating each match as a confrontation between two atomic entities, where each team possesses an estimated strength parameter derived from observed data. However, since clubs are formed by a collection of players, it is natural to consider individual-level modelling, where each player is treated as an atomic entity for the model. In this direction, [Wolf, Schmitt, and Schuller \(2020\)](#) proposed a player rating system based on the Elo algorithm, originally developed by [Elo \(1961\)](#) and still used today by the Fédération Internationale des Échecs (FIDE) for chess rankings. Since the Elo update rule corresponds to stochastic gradient ascent on the Bradley-Terry log-likelihood ([ALDOUS, 2017](#)), the work of [Wolf, Schmitt, and Schuller \(2020\)](#) can be viewed as a player-level extension of the Bradley-Terry

framework. Using data from 18 European competitions from seasons 2014/15 to 2017/18, they calculated player rankings and generated insights into each player’s impact on their club’s performance.

A critical aspect of statistical modelling is ensuring that the methods used for posterior inference correctly reproduce the intended posterior distributions. When this is not the case, the resulting parameter estimates and uncertainty quantification can be misleading, potentially leading to incorrect conclusions. Traditional diagnostics, such as trace plots and effective sample size, can detect some problems but may miss systematic biases. Nevertheless, they provide no direct guarantee that the samples are drawn from the correct posterior distribution.

Cook, Gelman, and Rubin (2006) introduced the concept of Simulation-Based Calibration (SBC) as a method for validating Bayesian inference procedures. The key insight is that if an inference procedure correctly reproduces the posterior distribution, then the rank of the true parameter value among posterior samples should follow a uniform distribution. This property provides a powerful diagnostic tool: systematic deviations from uniformity indicate that the inference procedure is not correctly reproducing the posterior distribution. Talts et al. (2020) formalized this approach and provided a comprehensive framework for applying SBC to validate Bayesian models. They showed that SBC can detect various types of problems, including bias, under-dispersion, and overdispersion in posterior distributions. More recently, Modrák et al. (2025) extended SBC to high-dimensional settings, by testing uniformity of test quantities.

In the context of Brazilian football, some initiatives include the *Chance de Gol* website (ARRUDA, 1999), which has been providing match outcome probabilities since 1999, and the *Probabilidades no Futebol* project (DEPARTAMENTO DE MATEMÁTICA, UFMG, 2005), maintained by the Department of Mathematics at the Universidade Federal de Minas Gerais (UFMG) since 2005. On the academic side, Diniz et al. (2018) compared several probabilistic predictive models, including Bayesian multinomial-Dirichlet formulations, evaluating their predictive performance on *Campeonato Brasileiro Série A* (also known as *Brasileirão*) matches using proper scoring rules and calibration diagnostics.

The present work makes three main contributions to the literature on Brazilian football modelling. First, a comprehensive dataset of Brazilian football matches spanning multiple seasons and competitions is introduced. Second, Simulation-Based Calibration is applied to validate Bayesian inference in football models. Finally, club-level formulations are extended to the player level, enabling a more granular analysis of team performance.

2 Data Extraction and Processing

This chapter details the methodology for extracting, organising and processing data from digital match reports made available by the *Confederação Brasileira de Futebol* (CBF).

The CBF maintains, since 2013, a repository with open access to match reports in PDF format. These reports can be accessed through CBF’s website (<https://www.cbf.com.br>) or using the following URL pattern: https://conteudo.cbf.com.br/sumulas/{year}/{championship_id}{game_id}se.pdf. The championship identifiers are shown in Table 1. The game ID is an incremental integer, one for each game and defined before the start of the championship, which may result in gaps when a game is postponed during the season.

Championship ID	Championship Name	Details
142	<i>Campeonato Brasileiro Série A</i>	Brazilian First Division
242	<i>Campeonato Brasileiro Série B</i>	Brazilian Second Division
342	<i>Campeonato Brasileiro Série C</i>	Brazilian Third Division
542	<i>Campeonato Brasileiro Série D</i>	Brazilian Fourth Division
424	<i>Copa do Brasil</i>	National Cup

Table 1 – **Championship ID pattern on CBF’s website.** The IDs used for accessing match reports on CBF’s website.

2.1 Structure of CBF PDF Reports

Figure 1 shows an example of a match report from the 2013 season between CR Vasco da Gama and Associação Portuguesa de Desportos. This example illustrates how the report is structured and helps understand how the principal information can be extracted.

The report begins with a header containing match details: championship, round, game, date, time, and stadium. Then, we have the referee information, followed by the chronological details of the game.

Next, we have the player line-up relation as two tables, one for each club. For each player, the table includes their jersey number, nickname, and full name. Players in the starting line-up are marked with a “T”, while substitutes are marked with a “R”. The table also indicates whether each player is professional (P) or amateur (A). Each player is identified by their unique CBF id, which we use internally throughout our data extraction and processing workflow instead of player names, since names are more sensitive

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Confederação Brasileira de Futebol

Publicação da Simbio: 2009/2013 21-23

No data via ZENODO 10.11

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Comissão Técnica					Refereencial IDP	
Técnicos:	Vasco & Gamal J.R.				Técnicos:	Edson Pinheiro, Bruno Pinho - 4552135
Auxiliar Técnico:	Rene Carlos Kneist Walter - 5229638 CNMSP				Auxiliar Técnico:	Geovani José Sobral - 381123276
Médico:	Maurício Teves Matar - 5229638 CNMSP				Médico:	Cláudio Lopes - 70005
Preparador Técnico:	Gisela Araújo Dos Santos - 020552021 FPEF				Preparador Técnico:	Yeniê Carolina De Almeida - 091016 GSP
Massagista:	Bruno Santos Queiroz Gomes - 48959 F CREDITO				Massagista:	Dirceu Ferreira De Lima - 28636658

Gols						
Tempo	32/31	Nº	Time	Nome do Jogador	Equipe	
02:00	21	11	NR	Carlos Viciente Tenório Médica	Vasco da Gama	

Cartões Amarelos						
Tempo	32/31	Nº	Nome do Jogador	Equipe		
03:00	31	7	Orlando Luis Sarto	Portuguesa		
			Motivo: Caráter adversário			
28:00	31	8	Victor Yoshimori Yutan Flores	Vasco da Gama		
			Motivo: Caráter adversário			
39:00	21	14	Luiz Carlos Teixeira	Vasco da Gama		
			Motivo: Retenção e retenção da partida			

Cartões Vermelhos						
Tempo	32/31	Nº	Nome do Jogador	Equipe		
20:00	21	7	Orlando Luis Sarto - Associação Portuguesa de Desportos			
Descrição: Aos 23 minutos do 2º tempo, expulsão do campo de jogo em decorrência do 2º cartão amarelo o atleta sr. Orlando Luis Sarto nº 7 do time português por ter levado um jogador adversário ao chão por falta no ar, sendo penalizado máximo o cartão nº 27 do mesmo, segundo as regras da FPF, sendo esse considerado o último do caso por sendo jogador com a bola depois de fazer o placar de 1 a 0 com o primeiro golpear adversário, sendo esse ocorrido na lateralidade do campo de jogo do lado da seta da jogadora expulsoa informado via rádio e após expulso a banca não aderindo com o cartão amarelo em jogada anterior por também ter colocado o seu adversário no chão de bola, onde houve ocorrência de atendimento médico ao atleta atingido e seus contatos na partida até o seu fim sem outras ocorrências a serem relatadas.						
2º Cartão Amarelo						

Confederação Brasileira de Futebol

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Entwickelt durch: 2309/2013 14.3

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Ocorrências / Observações			
Nada houve e anormal			
Observações Eventuais			
NADA HOUVE DE ANORMAL			

Substituições				
Tempo	TIME	Equipe	Entrou	Saiu
17:00	2T	Vasco da Gama	19 - Edmilson dos Santos Silva	27 - Dalcson Soares de Silva
21:00	2T	Portuguesa	17 - Henrique Campos Santos	9 - Luiz Fernando Borges Roma
21:00	2T	Portuguesa	16 - Moises Lima Magalhães	8 - Carlos Rodrigo Correa
24:00	2T	Vasco da Gama	22 - Jucimar Lima Pischke	21 - Felipe Ramos Ighez Basto
27:00	3T	Vasco da Gama	17 - Wendel Geraldo Maurício E ...	27 - Alexson Euter de Freitas
37:00	2T	Portuguesa	18 - Juan Jose Araya	10 - William de Souza Silva

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FIGURE 11.3

Figure 1 – **Example of a CBF match report.** Example report from a match between CR Vasco da Gama and Associação Portuguesa de Desportos, which took place on May 25th, 2013, for the first round of the Brazilian First Division (*Série A*).

to encoding issues and may have inconsistent accentuation across reports. Following the players' relation we have information on the coaching and medical staff.

After the team information, the report contains data on the events that occurred during the match. This section starts with the goals scored during the match. The report saves this information on two columns for time because in Brazil the game is divided and counted in two periods with 45 minutes each. In this sense, unlike other countries, the minute 65 of a game is assigned as 20 minutes of the second period. For each goal, the table includes when it was scored (minute and period), the scorer's jersey number and name, the type of goal, which can be open-play (NR), penalty (PN), own goal (CT), or free kick (FT), and the club for which the goal counts.

Yellow and red card data is found right after the goal data. The first section is related to yellow cards, while the second is to red cards. The organization of these sections follows the same structure as the goals table, except that the goal type field is not applicable. For each card, the table includes when it was shown (minute and period), the player's jersey number and name. An extra field, right below the player's name, indicates the reason for the card, though this information is not extracted in our retrieval process.

After these sections, we have substitution information. This section is also organized as a table. For each substitution, the table includes when it was made (minute and period), which club made the substitution, and the jersey number and full name of both the player entering and the player exiting the field.

At the end of the report, we have two sections for any occurrence or observation that the referees' team thinks is relevant.

2.2 Data Retrieval Process

The data retrieval process begins with extracting the PDF reports from CBF's website. We use the Python package `REQUESTS` (REITZ, KENNETH AND CONTRIBUTORS, 2024), passing the URL pattern described in Section 2.1 as an argument. After making the request, we validate the response and, if valid, save the PDF file to disk. To avoid overwhelming CBF's server, once a report is successfully saved, we do not make additional requests for that same report.

With the saved PDF, we open it using `PDFREADER` from the `PYPDF2` package (PYPDF2 DEVELOPMENT TEAM, 2024) and extract all text content to a CSV file, which is then saved to disk. The parsing process is implemented in Python using the `RE` package (PYTHON SOFTWARE FOUNDATION, 2024) for regular expression matching to extract and parse the data from the CSV files, saving the structured information into JSON files.

From the file header, we extract the clubs' names and the final match result. For both clubs in the game, we collect information about all players who participated in that match, including their jersey number, nickname and full name, starting condition (in the line-up or on the bench), professional status, the player's CBF code, and the club represented by the player in that game.

After extracting player information, we collect data on goals scored in the match. For each goal, we save the moment (minute and period), the goal type (open-play, penalty, free kick, or own goal), the jersey number and name of the scorer, and the club for which the goal was scored.

We also extract information about substitutions that occurred during the game. For each substitution, we save the moment of the change, the club that made the substitution, and the jersey number and name of both the player who entered and the player who exited.

Finally, we record information about yellow and red cards. This data is saved with the same structure as goals, except for the goal type field. An important note is that in this process, each half of the game is clipped to 45 minutes. Therefore, all events that occurred during injury time are recorded as events at minute 45 of the corresponding period.

2.3 Data Treatment and Normalization

As discussed in the previous section, most of the data registered in the reports concerns a player's jersey number and name. This poses two problems: first, some clubs adopt a variable jersey number distribution during the season, where players in the starting line-up always have numbers from 1 to 11. Secondly, using player names can be problematic because we have both the player's nickname and "full name", but the full name is often incomplete or truncated. To address these issues, we convert all player references to CBF's unique ID system. During the data extraction process, we substitute the player's name and jersey number with their personal CBF ID, ensuring consistent identification across matches and seasons.

Furthermore, our data processing workflow segments each game using all player substitutions and red cards to create temporal segments. For each game, we generate "sub-games", each representing a period during which the same set of players was on the pitch, bounded by substitutions or red card events. The final result is a JSON file with all games structured as shown in Figure 3.

```
{
  "001": {
    "Summary": {
      "Home": "Vasco da Gama / RJ",
```

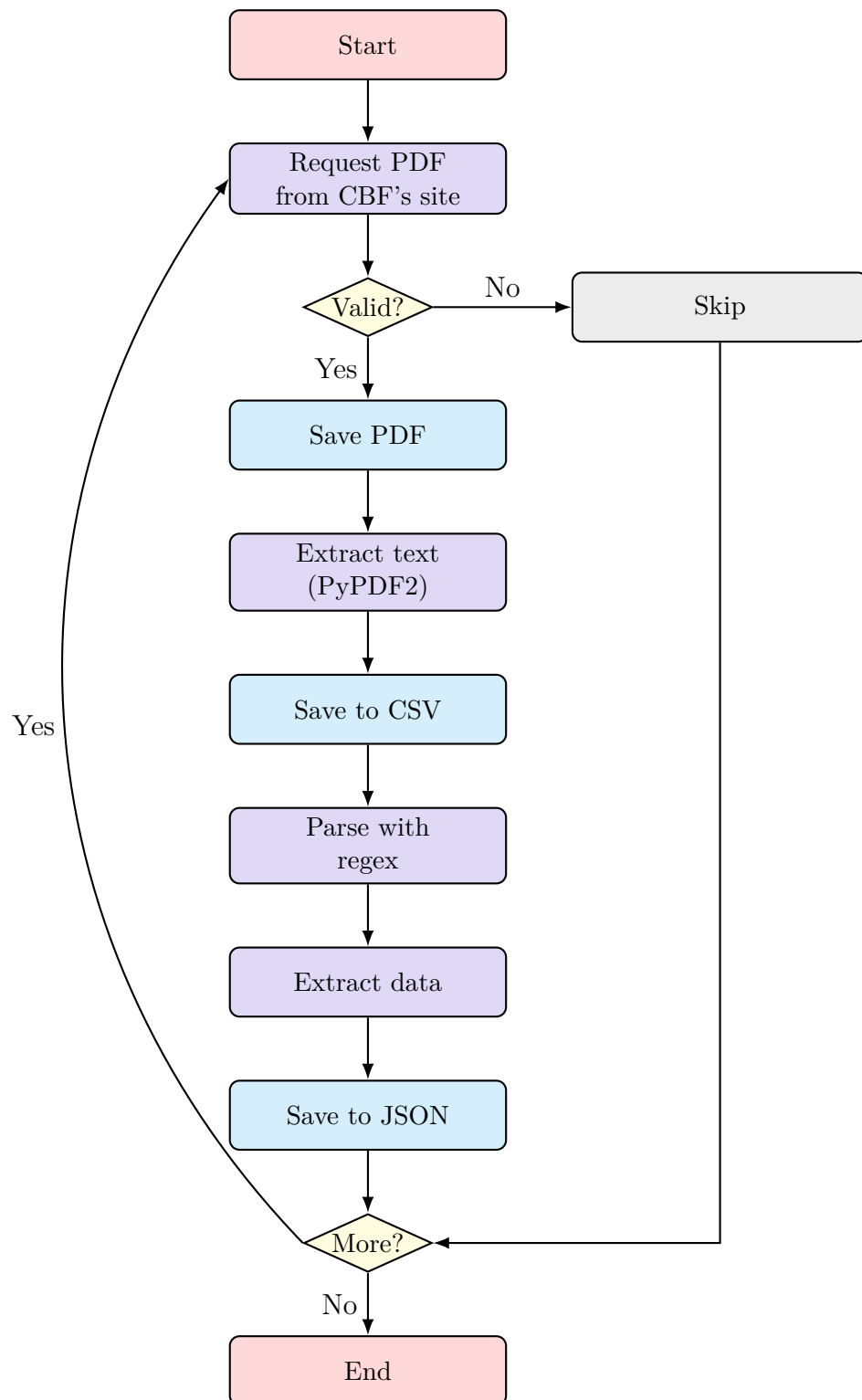


Figure 2 – **Data extraction workflow.** Overview of the extraction process: first extracting the reports from CBF's website, then processing the information through PDF to CSV conversion and JSON parsing.


```

    "Away": "Portuguesa / SP",
    "Result": "1 X 0",
    "Date": "25/05/2013",
    "Time": "18:30",
    "Stadium": "São Januário / Rio de Janeiro"
  },
  "0": {
    "Home": {
      "Squad": ["146379", "186825", "430895", "165075", "402413", "307406",
        "357424", "317751", "156194", "167111", "175011"],
      "Cards": [[[28, False], "430895"]],
      "Goals": [[[47, False], "402413"]]
    },
    "Away": {
      "Squad": ["159483", "158204", "160789", "134212", "151104", "160357",
        "176908", "138048", "190258", "136911", "188158"],
      "Cards": [[[23, False], "176908"]],
      "Goals": []
    },
    "Time": 62
  },
  "1": {
    "Home": {
      "Squad": ["146379", "186825", "430895", "165075", "402413", "307406",
        "357424", "317751", "156194", "167111", "143841"],
      "Cards": [],
      "Goals": []
    },
    "Away": {
      "Squad": ["159483", "158204", "160789", "134212", "151104", "160357",
        "176908", "138048", "190258", "136911", "188158"],
      "Cards": [],
      "Goals": []
    },
    "Time": 4
  }
}

```

Figure 3 – **Final JSON structure example.** Example of the game between CR Vasco da Gama and Associação Portuguesa de Desportos, showing two sub-games triggered by a substitution (CBF ID 175011 replaced by 143841) at the 62nd minute. The “Summary” field stores general metadata such as team names, final result, date/time, and venue.

The JSON structure can be interpreted as follows:

- "001": Represents the unique identifier for the match.
 - "Summary": Provides a summary of the match, including home team, away team, final result, date, time, and stadium.
 - Other second-level keys (e.g., "0", "1", "2", ...) represent different segments of the match. These segments are created whenever a substitution or red card occurs, even if the red card is shown to a reserve player.
 - * "Home" and "Away": Information about each team during this segment.
 - "Squad": List of player CBF IDs for the team.
 - "Cards": List of cards received by team players. Each card is represented by an array where the first element is a tuple with the minute the card was shown and a boolean flag indicating whether it occurred during injury time of the current segment. The second element is the player's CBF ID.
 - "Goals": List of goals scored by the team, each represented by an array where the first element is a tuple with the minute the goal was scored and a boolean flag indicating whether it occurred during injury time. The second element is the scorer's CBF ID.
 - * "Time": Duration of this segment in minutes.

An important consideration in this process is that all reports are written manually, which means they may contain typos or mistakes. When we identified anomalies that caused errors in the retrieval process, we manually corrected them. However, this correction process was only performed for First Division (*Série A*) matches. Consequently, in lower divisions, we can expect more anomalies that were not treated, which may result in potentially inferior data quality for those competitions.

2.4 Dataset Characterization

The final dataset covers match reports from 2013 to 2025, encompassing the main Brazilian football competitions: the four divisions (*Série A*, *Série B*, *Série C*, and *Série D*) and the national cup (*Copa do Brasil*). In total, the dataset maps 16,800 unique players who played for 368 unique clubs across all competitions. The complete dataset is publicly available on Zenodo ([MICHELS; CARVALHO, 2026](#)).

Table 2 shows the number of matches by season and championship, while Tables 3 and 4 show the distribution of clubs and players. In these tables, each value corresponding to both a season and a championship represents the total number of games, unique players, or unique clubs for that championship during that season. The “Total unique” columns in Tables 3 and 4 aggregate the players/clubs across all championships for the corresponding

season, while the “Total unique” rows aggregate across all seasons for a given championship. The final value in the bottom-right corner shows the total number of unique players/clubs in the entire dataset.

Season	<i>Série A</i>	<i>Série B</i>	<i>Série C</i>	<i>Série D</i>	<i>CdB</i>	Total games
2013	380	380	213	188	156	1,317
2014	380	379	194	195	159	1,307
2015	380	380	194	190	158	1,302
2016	379	380	194	266	162	1,381
2017	380	380	193	266	120	1,339
2018	380	380	194	266	120	1,340
2019	380	379	194	266	120	1,339
2020	380	379	206	516	120	1,601
2021	380	380	206	518	122	1,606
2022	380	380	216	510	122	1,608
2023	380	380	216	510	122	1,608
2024	380	379	216	508	122	1,605
2025	380	375	214	507	121	1,597
Total	4,939	4,931	2,650	4,706	1,724	18,950

Table 2 – **Total number of games by season and championship.** “CdB” stands for *Copa do Brasil*, the national cup competition.

Season	<i>Série A</i>	<i>Série B</i>	<i>Série C</i>	<i>Série D</i>	CdB	Total unique clubs
2013	20	21	21	40	88	130
2014	20	20	20	41	87	132
2015	20	20	20	40	87	125
2016	20	20	20	68	86	147
2017	20	20	20	68	91	134
2018	20	20	20	69	91	138
2019	20	20	20	68	91	131
2020	20	20	20	68	91	136
2021	20	20	20	69	92	134
2022	20	20	20	64	92	129
2023	20	20	20	66	93	133
2024	20	20	20	65	92	137
2025	20	20	20	64	93	137
Total unique clubs	37	66	83	297	294	368

Table 3 – **Total number of clubs by season and championship.** “CdB” stands for *Copa do Brasil*, the national cup competition.

Season	<i>Série A</i>	<i>Série B</i>	<i>Série C</i>	<i>Série D</i>	<i>CdB</i>	Total unique players
2013	709	811	687	1,061	1,799	3,644
2014	736	833	650	1,043	1,872	3,671
2015	715	805	667	1,056	1,897	3,672
2016	722	775	666	1,603	1,847	4,012
2017	722	731	601	1,551	1,591	3,783
2018	720	740	584	1,563	1,605	3,745
2019	691	740	618	1,553	1,579	3,748
2020	733	791	690	2,032	1,695	4,381
2021	713	733	674	2,049	1,851	4,398
2022	747	796	685	2,037	1,835	4,322
2023	736	771	682	1,886	1,833	4,127
2024	709	756	674	1,965	1,817	4,231
2025	731	786	693	1,973	1,774	4,278
Total unique players	3,467	4,657	4,738	11,141	9,782	16,800

Table 4 – **Total number of players by season and championship.** The totals do not correspond to simple sums because they count unique players across seasons.

3 Simulation-Based Calibration

Simulation-Based Calibration (SBC) is a methodology designed to assess the calibration and correctness of Bayesian inference algorithms. It was introduced by [Cook, Gelman, and Rubin \(2006\)](#) and formalized by [Talts et al. \(2020\)](#) as an approach to diagnose whether a probabilistic model, together with its inference engine, correctly reproduces the intended posterior distribution. Unlike traditional model checking or convergence diagnostics, SBC focuses on the *calibration* of the entire inference process, helping to determine whether an inference algorithm behaves as expected when the model is correctly specified.

3.1 Theoretical Foundation

The main idea of SBC is to use the rank statistic as a diagnostic measure. For each simulation $i = 1, \dots, S$:

Algorithm 1: Single iteration of the SBC procedure. For each simulation, sample parameters from the prior, generate synthetic data, run inference, and compute the rank statistic.

- 1 Sample $\theta_i \sim p(\theta)$;
 - 2 Sample $y_i \sim p(y \mid \theta_i)$;
 - 3 Obtain posterior samples $\{\theta_i^{(1)}, \dots, \theta_i^{(L)}\} \leftarrow \mathcal{A}(y_i)$;
 - 4 Compute rank R_i of θ_i among the posterior samples;
-

Under the assumption that the inference algorithm, $\mathcal{A}$, correctly reproduces the posterior distribution $p(\theta \mid y)$, the rank statistic follows a discrete uniform distribution on $\{0, 1, \dots, L\}$. Theorem 1 states and proves this result through an approach similar to the argument in Appendix B of [Talts et al. \(2020\)](#).

Theorem 1 (Rank Uniformity Under Correct Posterior Inference). *If the inference algorithm $\mathcal{A}$ used in Algorithm 1 correctly reproduces the posterior distribution $p(\theta \mid y)$ for all θ and y , then the rank statistic R_i produced by the SBC iteration is distributed uniformly on $\{0, 1, \dots, L\}$.*

Proof. For simplicity, we will omit the indices in this proof. Now, without loss of generality, we can relabel all the posterior samples such that

$$\theta^{(1)} \leq \theta^{(2)} \leq \dots \leq \theta^{(L)}.$$

Then, we can write the probability mass function of the rank statistic as

$$\begin{aligned} p(r) &= \binom{L}{r} \int p(y, \theta) \cdot \left(\prod_{l=1}^r \int_{-\infty}^{\theta} p(\theta^{(l)} | \theta, y) d\theta^{(l)} \right) \cdot \left(\prod_{l=r+1}^L \int_{\theta}^{\infty} p(\theta^{(l)} | \theta, y) d\theta^{(l)} \right) d\theta dy \\ &= \binom{L}{r} \int p(y, \theta) \cdot \left(\int_{-\infty}^{\theta} p(\theta^{(l)} | \theta, y) d\theta^{(l)} \right)^r \cdot \left(1 - \int_{-\infty}^{\theta} p(\theta^{(l)} | \theta, y) d\theta^{(l)} \right)^{L-r} d\theta dy. \end{aligned}$$

Since we condition on the data y , the posterior sample is independent of the generative parameter θ , so

$$p(\theta^{(l)} | \theta, y) = p(\theta^{(l)} | y).$$

Also, using that $p(y, \theta) = p(\theta | y) \cdot p(y)$ we can rewrite

$$\begin{aligned} p(r) &= \binom{L}{r} \int p(y, \theta) \cdot \left(\int_{-\infty}^{\theta} p(\theta^{(l)} | \theta, y) d\theta^{(l)} \right)^r \cdot \left(1 - \int_{-\infty}^{\theta} p(\theta^{(l)} | \theta, y) d\theta^{(l)} \right)^{L-r} d\theta dy \\ &= \binom{L}{r} \int p(y) \int p(\theta | y) \cdot \left(\int_{-\infty}^{\theta} p(\theta^{(l)} | y) d\theta^{(l)} \right)^r \cdot \left(1 - \int_{-\infty}^{\theta} p(\theta^{(l)} | y) d\theta^{(l)} \right)^{L-r} d\theta dy. \end{aligned}$$

Since the algorithm correctly reproduces the posterior distribution, we have

$$p(\theta^{(l)} | y) = p(\theta | y).$$

Now, define

$$u = \int_{-\infty}^{\theta} p(t | y) dt = F_y(\theta) \implies u \in [0, 1] \wedge du = p(\theta | y) d\theta,$$

which allows us to change the variables in the integral. Then

$$\begin{aligned} p(r) &= \binom{L}{r} \int p(y) \cdot \int_0^1 u^r \cdot (1-u)^{L-r} du dy \\ &= \frac{L!}{r! \cdot (L-r)!} \int p(y) \cdot \frac{r! \cdot (L-r)!}{(L+1)!} dy \\ &= \frac{1}{L+1} \int p(y) dy \\ &= \frac{1}{L+1}, \end{aligned}$$

which is consistent with a uniform distribution over the $L+1$ possible ranks of the set $\{0, 1, \dots, L\}$. $\square$

In practice, Theorem 1 states that by repeating this process across S simulations, we can plot and see visually or test statistically whether the computed ranks are uniformly distributed to validate the calibration of the inference process. Figure 4 shows the workflow of this validation.

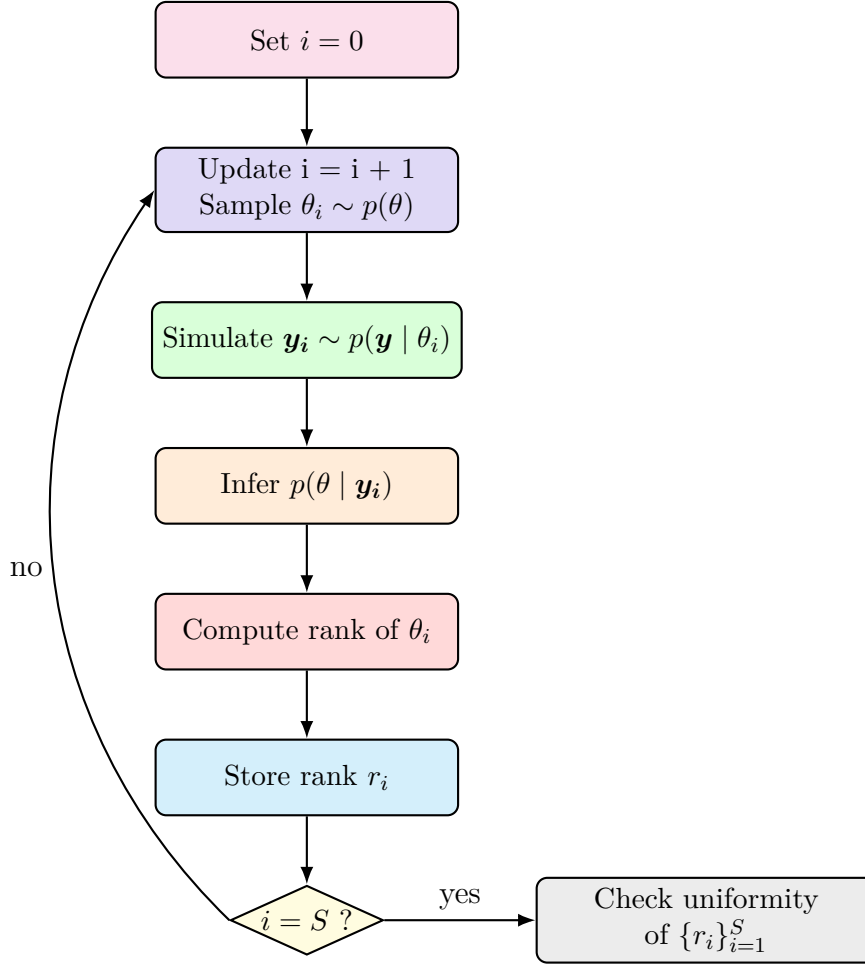


Figure 4 – **SBC workflow**. For each simulated dataset, we generate θ_i , simulate data y_i , infer the posterior, and check whether ranks of θ_i are uniformly distributed.

3.2 Uniformity and Deviations

As mentioned in the previous section, if the inference algorithm is unbiased and correctly implemented, the histogram of ranks should be approximately flat. Bias and under/overdispersion can be identified looking into the histogram, Empirical Cumulative Distribution Function (ECDF) or ECDF difference plots of the computed ranks. Figure 5 shows how common deviations can be identified in the ECDF difference plot. For this example, we ran 2560 simulations, each generating $\mu \sim \mathcal{N}(0, 1)$ and sampling 10 points from a distribution $\mathcal{N}(\mu, 1)$. The inference process ran 100 iterations of warm-up and 127 iterations with posterior sampling.

As we can see in Figure 5, a bias in the posterior reflects on a U-shape or an inverted U-shape in the ECDF differences plot. On the other hand, under/overdispersion in the posterior results in a S-shape.

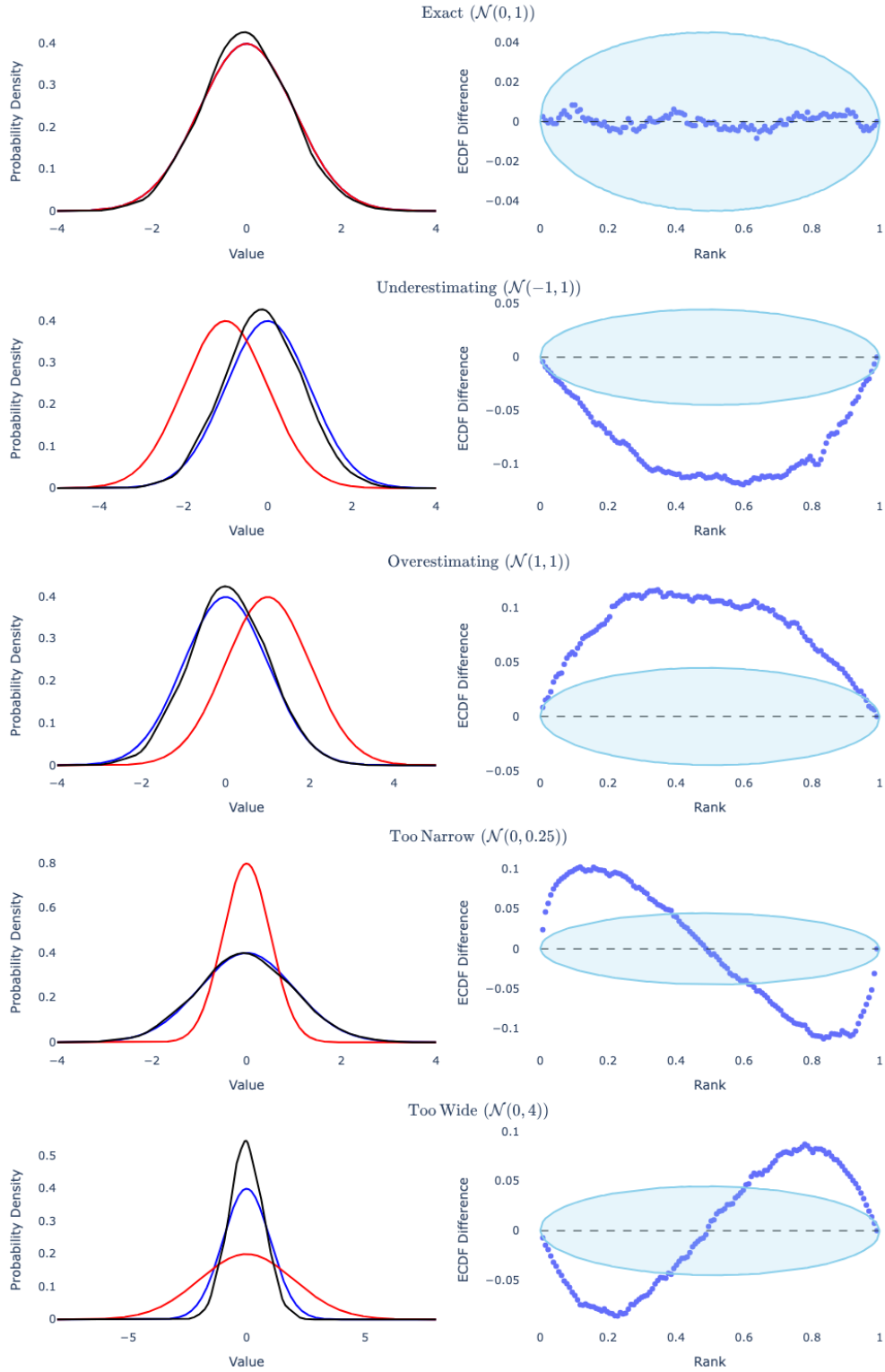


Figure 5 – **SBC applied on synthetic data.** The true μ distribution is given by the blue distribution. The red distribution was set as data conditional distribution on inference process during the SBC validation. The black distribution is the kernel density estimate (KDE) for the μ posterior. The second column shows the ECDF difference plot for the normalized rank distribution. Under a well calibrated model, this ECDF is expected to be a straight line from $(0, 0)$ to $(1, 0)$.

3.3 Computational and High Dimensional Considerations

As illustrated in Figure 4, SBC can be computationally expensive, as it requires repeated posterior inference for many synthetic datasets. This cost can be mitigated through parallelisation, since each inference run is independent. Despite the high computational cost, performing SBC early in the modelling process allows for detection of coding mistakes or conceptual errors before substantial resources are spent on large-scale analyses. For this reason, it is recommended that SBC be considered as part of the Bayesian model development process, rather than an afterthought, as emphasised by [Gelman et al. \(2020\)](#).

These computational challenges become even more pronounced in high-dimensional models, where a large number of parameters must be assessed simultaneously. Applying SBC independently to each parameter is conceptually straightforward but has a huge cost. To reduce this cost, the workflow in Figure 4 can be extended to high dimensional models by just taking θ as a d -dimensional vector. For each component of θ , rank statistics can be computed and stored across simulations, and their empirical distributions can then be checked for uniformity. While this approach increases storage and computational demands, it provides a systematic way to diagnose inference quality even in complex, high-dimensional settings.

3.4 Experimental Design

All SBC experiments were designed to represent the typical format of a *Brasileirão Série A* season. For each model, $S = 500$ synthetic datasets were generated, each comprising $n = 20$ teams competing in a double round-robin tournament with 380 matches.

The inference process uses Stan ([CARPENTER et al., 2017](#)) via the CmdStanPy interface ([STAN DEVELOPMENT TEAM, 2023](#)). Each simulation was run with 4 parallel chains, 1,000 warm-up iterations, and 1,000 sampling iterations per chain, yielding 4,000 posterior samples per simulation. The sampler was configured with an adaptation target acceptance probability of $\delta = 0.95$ and a maximum tree depth of 10.

For each simulation i , the true parameter values θ_i were sampled from the prior distributions specified in the model. Synthetic data $\mathbf{y}_i$ were generated according to the likelihood, and posterior samples $\{\theta_i^{(1)}, \dots, \theta_i^{(L)}\}$ were obtained via Markov Chain Monte Carlo (MCMC). The normalized rank of each true parameter value among its posterior samples was recorded. Under correct posterior inference, these ranks should follow a uniform distribution on $[0, 1]$, as established in Theorem 1.

Since there are a lot of models, with many parameters, to evaluate, we use a systematic approach to identify potential miscalibrations. For each parameter in a model, we compute the ECDF difference plot and check whether any point falls outside the 95%

confidence envelope. If at least one point lies outside this envelope, that parameter is flagged, and we record the total number of points outside the envelope for that parameter. Aggregating across all parameters of a model, we obtain two summary statistics: the total number of flagged parameters (“Flags”) and the cumulative count of points outside the confidence envelopes (“Points Out”). A model with many flagged parameters or a large cumulative count suggests potential miscalibration. However, under correct calibration, we expect approximately 5% of parameters to be flagged by chance alone, and minimal deviations may still occur due to the inherent randomness of the simulations.

4 Methodology

This chapter presents a collection of probabilistic models used to describe football match outcomes. Rather than introducing each model independently, we adopt a unified approach in which both model families are first defined through general formulations. Specific models are then obtained by imposing structural constraints and simplifying assumptions on these general representations.

We begin with the Bradley–Terry family of models, which describe match outcomes in terms of latent team skill parameters. A general formulation is introduced that encompasses home effects and the possibility of draws, from which simpler variants naturally emerge by restricting the corresponding parameters. This formulation provides a common reference point for interpretation, comparison, and validation across the different Bradley–Terry specifications.

We then consider Poisson-based models, which explicitly model the number of goals scored by each team. The general Poisson formulation incorporates offensive and defensive effects, home effects, and potential dependence between teams’ scoring processes. As with the Bradley–Terry case, the specific Poisson models used in this work are obtained through systematic simplifications of the general formulation, resulting in a structured hierarchy of models with increasing complexity.

All models are intentionally kept simple, both in structure and in the number of parameters. This design choice reflects our goal of validating the models using Simulation-Based Calibration and fitting them to real data from the Brazilian First Division. The predictive performance of the models is compared, and the best-performing specifications are subsequently extended to player-level versions, where the dimensionality of the parameter space increases substantially. Maintaining simple club-level formulations is therefore essential to control the computational cost of these extensions while preserving interpretability.

The chapter also describes how the club-level formulations are adapted to player-level models, outlining the structural modifications required to replace team-level to player-level parameters. The inference procedure is then detailed, covering the prior distributions, the data inputs at each level, and the MCMC sampling configuration. Finally, the chapter concludes with the evaluation methodology used to assess the predictive performance of the models on real data, including the prediction scheme, baseline predictors, and proper scoring rules.

4.1 Bradley-Terry model

Bradley–Terry models (BRADLEY; TERRY, 1952) provide a natural starting point for modelling football match outcomes, as they directly relate the probability of winning to latent team skill parameters. In their simplest form, these models treat matches as binary events, focusing only on the final outcome rather than the underlying scoreline.

In this work, we consider a family of Bradley–Terry–type models of increasing complexity, beginning with a basic binary formulation and progressively incorporating home effect and the possibility of draws. These models serve both as a baseline for predictive performance and as a test for methodological validation, given their relatively low-dimensional parameter space and well-understood identifiability properties.

4.1.1 General Formulation

Before presenting the specific variants of the Bradley–Terry model, we introduce a general formulation that incorporates all its structural components: team skills, home effect, and the possibility of draws. This serves as a conceptual starting point, from which different models naturally emerge through parameter restrictions and simplifying assumptions.

The Bradley–Terry model can be expressed in terms of team strengths and outcome probabilities. For a match between team i (home) and team j (away), the strengths are defined as:

$$s_i = \exp(\theta_i + \nu) \quad (4.1)$$

$$s_j = \exp(\theta_j), \quad (4.2)$$

where:

- $\theta_i \in \mathbb{R}$ is the log-skill parameter for team i ;
- $\nu \in \mathbb{R}$ is the home effect parameter;
- s_i is the strength of team i when playing at home;
- s_j is the strength of team j when playing away.

The probability of each outcome is then given by:

$$P(\text{team } i \text{ wins}) = \frac{s_i}{s_i + s_j + s_{\text{tie}}} \quad (4.3)$$

$$P(\text{team } j \text{ wins}) = \frac{s_j}{s_i + s_j + s_{\text{tie}}} \quad (4.4)$$

$$P(\text{draw}) = \frac{s_{\text{tie}}}{s_i + s_j + s_{\text{tie}}}, \quad (4.5)$$

where $s_{\text{tie}} = \kappa\sqrt{s_i \cdot s_j}$ is the tie strength, with $\kappa \geq 0$ being the tie parameter, following the approach of Davidson (1970). When $\kappa = 0$, draws are not allowed and the model reduces to a binary outcome model, with the outcome $y_{i,j}$ following a Bernoulli distribution. When $\kappa > 0$, the outcome $y_{i,j}$ follows a categorical distribution with these probabilities.

Given a set of matches $\mathcal{M}$, the likelihood can be computed as:

$$\mathcal{L}(\boldsymbol{\theta}, \nu, \kappa; \mathbf{y}) = \prod_{m \in \mathcal{M}} P(y_m \mid \theta_{i_m}, \theta_{j_m}, \nu, \kappa), \quad (4.6)$$

where y_m is the outcome of match m between teams i_m and j_m .

The Bradley-Terry model defined by equations 4.3, 4.4, and 4.5 presents identifiability issues, with the likelihood remaining invariant under translations of all log-skill parameters by a constant c :

$$\mathcal{L}(\boldsymbol{\theta}, \nu, \kappa; \mathbf{y}) = \mathcal{L}(\boldsymbol{\theta} + c, \nu, \kappa; \mathbf{y}), \quad (4.7)$$

since:

$$\begin{aligned} (\theta_i + c) + \nu - (\theta_j + c) &= \theta_i + \nu - \theta_j, \\ \exp(\theta_i + c + \nu) &= \exp(c) \cdot \exp(\theta_i + \nu), \\ \exp(\theta_j + c) &= \exp(c) \cdot \exp(\theta_j), \end{aligned}$$

and the constant $\exp(c)$ cancels out in the ratio of strengths.

To ensure model identifiability, we impose a sum-to-zero constraint on the log-skill parameters:

$$\sum_{i=1}^n \theta_i = 0. \quad (4.8)$$

This constraint is implemented by parametrizing the vector as:

$$\boldsymbol{\theta} = \left[\theta_1, \theta_2, \dots, \theta_{n-1}, -\sum_{i=1}^{n-1} \theta_i \right], \quad (4.9)$$

where only the first $n - 1$ parameters are estimated, and the last is determined by the constraint.

4.1.2 Model Variants

Different models arise from simplifying assumptions about the parameters in the general formulation. The models can be categorized based on two key dimensions:

1. **Home Effect:** Present ($\nu \neq 0$) vs. Absent ($\nu = 0$);
2. **Ties Allowed:** Present ($\kappa > 0$) vs. Absent ($\kappa = 0$).

This section describes the four Bradley-Terry model variants (BT1–BT4) used to model match outcomes in football matches in the present work. These models are organized from the simplest binary outcome model without home effect, BT1, to the most complex ternary outcome model with home effect, BT4. The model names were chosen for simplicity, avoiding unnecessary repetition.

Model **BT1** is the simplest Bradley-Terry model, using a single log-skill parameter θ_i per team with no home effect and no ties allowed. The log-odds of team i winning against team j is given by $\log(\text{odds}) = \theta_i - \theta_j$, which corresponds to a binary outcome model where $P(\text{team } i \text{ wins}) = \exp(\theta_i) / (\exp(\theta_i) + \exp(\theta_j))$.

Model **BT2** extends BT1 by incorporating a home effect parameter ν . The log-odds becomes $\log(\text{odds}) = \theta_i + \nu - \theta_j$ for the home team, capturing the additional effect of playing at home. Like BT1, this model does not allow ties.

Model **BT3** extends BT1 by allowing ties through the introduction of a tie parameter $\kappa \geq 0$. The tie strength is defined as $s_{\text{tie}} = \kappa \sqrt{s_i \cdot s_j}$, where $s_i = \exp(\theta_i)$ and $s_j = \exp(\theta_j)$. This model does not include home effect, so the strengths are symmetric with respect to home and away contexts.

Model **BT4** combines the features of BT2 and BT3, incorporating both home effect and ties. The home team's strength becomes $s_i = \exp(\theta_i + \nu)$, while the away team's strength remains $s_j = \exp(\theta_j)$. The tie strength is $s_{\text{tie}} = \kappa \sqrt{s_i \cdot s_j}$, providing the most flexible specification that accounts for both home effect and the possibility of draws.

The taxonomy of these model variants is illustrated in Figure 6.

4.2 Poisson model

From a modelling perspective, Bradley-Terry models treat matches as binary or ternary events by collapsing the game score into a win/draw/loss indicator. This abstraction makes it impossible for the model to distinguish between a dominant win and a narrow one, resulting in a loss of information about team performance.

In contrast, Poisson models (MAHER, 1982; DIXON; COLES, 1997; LEE, 1997; KARLIS; NTZOUFRAS, 2000) offer an alternative by modelling the home and away goals directly. This not only doubles the number of observed data per match, from an indicator to two count variables, but also allows the model to capture potential asymmetries between teams' performance.

4.2.1 General Formulation

The general formulation of the Poisson models presented in this section describes the most comprehensive structure considered in this work for modelling the number of

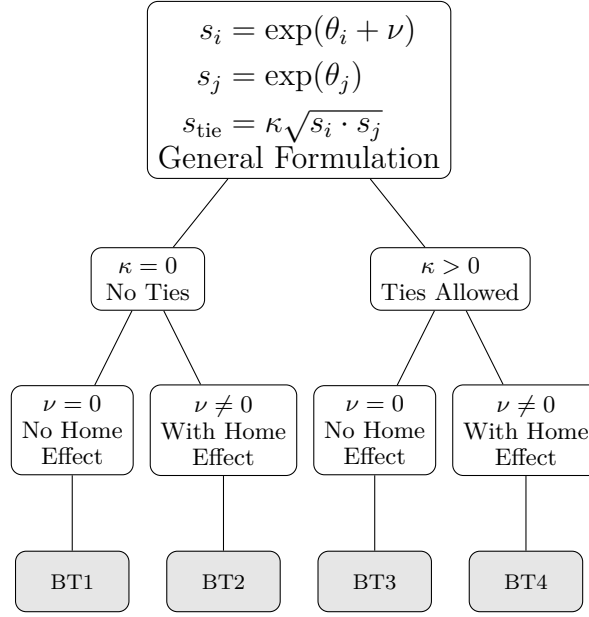


Figure 6 – **Taxonomy of the four Bradley-Terry model variants.** The root node displays the complete specification with log-skill parameters (θ_i), home effect (ν), and tie parameter (κ). The first level of branching corresponds to whether ties are allowed ($\kappa > 0$) or not ($\kappa = 0$). The second level distinguishes models with ($\nu \neq 0$) and without ($\nu = 0$) home effect. Each leaf node represents a specific model variant, with the model identifier (BT1–BT4) indicating the final configuration after all simplifications have been applied.

goals scored by each team. This parametrization includes distinct offensive and defensive effects, distinguishes between home and away contexts, and allows for the inclusion of a common effect term on both goal intensities, serving as the foundation for all model variants introduced later.

The general formulation for the log-intensity parameters can be expressed as:

$$\log(\lambda_{i,j,h}) = \alpha_i + \beta_j + \nu + \eta \quad (4.10)$$

$$\log(\lambda_{j,i,a}) = \delta_j + \gamma_i + \eta, \quad (4.11)$$

where:

- $\lambda_{i,j,h}$ is the expected number of goals scored by team i (home) against team j (away);
- $\lambda_{j,i,a}$ is the expected number of goals scored by team j (away) against team i (home);
- α_i is the offensive ability of team i when playing at home;
- γ_i is the defensive ability of team i when playing at home;
- β_j is the defensive ability of team j when playing away;
- δ_j is the offensive ability of team j when playing away;

- $\nu \in \mathbb{R}$ is the home effect parameter;
- $\eta \in \mathbb{R}$ is a common effect parameter that affects both teams equally.

The goals are then modelled as conditionally independent Poisson random variables:

$$g_{i,j} \mid \lambda_{i,j} \sim \text{Poisson}(\lambda_{i,j}) \quad (4.12)$$

$$g_{j,i} \mid \lambda_{j,i} \sim \text{Poisson}(\lambda_{j,i}), \quad (4.13)$$

where $g_{i,j}$ and $g_{j,i}$ are the goals scored by teams i and j respectively.

Given a set of matches $\mathcal{M}$, with home goals G_h and away goals G_a , the likelihood can be computed as:

$$\mathcal{L}(\boldsymbol{\theta}; G_h, G_a) = \prod_{m \in \mathcal{M}} \frac{\lambda_{m_h}^{g_{h_m}} \cdot \exp(-\lambda_{m_h})}{g_{h_m}!} \cdot \frac{\lambda_{m_a}^{g_{a_m}} \cdot \exp(-\lambda_{m_a})}{g_{a_m}!}, \quad (4.14)$$

where g_{h_m} and g_{a_m} are the total goals of home and away teams in match m , respectively, and λ_{m_h} and λ_{m_a} are the expected number of goals for the home and away teams in match m .

The general formulation presented in equations 4.10 and 4.11 presents identifiability issues. Specifically, the likelihood remains invariant under certain transformations of the parameters.

When $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, $\boldsymbol{\gamma}$, and $\boldsymbol{\delta}$ are all independent, adding a constant c to all attack parameters and subtracting the same constant from the defence parameters, while keeping ν and η unchanged, does not change the likelihood:

$$\mathcal{L}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\delta}, \nu, \eta; G_h, G_a) = \mathcal{L}(\boldsymbol{\alpha} + c, \boldsymbol{\beta} - c, \boldsymbol{\gamma} - c, \boldsymbol{\delta} + c, \nu, \eta; G_h, G_a). \quad (4.15)$$

This occurs because:

$$\begin{aligned} (\alpha_i + c) + (\beta_j - c) + \nu + \eta &= \alpha_i + \beta_j + \nu + \eta \\ (\delta_j + c) + (\gamma_i - c) + \eta &= \delta_j + \gamma_i + \eta. \end{aligned}$$

The constant c cancels out in the additions and subtractions, leaving the log-intensity parameters unchanged, and thus the likelihood remains invariant. Similarly, adding a constant c on η and subtracting c on both $\boldsymbol{\alpha}$ and $\boldsymbol{\delta}$ or both $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ keep the likelihood unchanged:

$$\begin{aligned} \mathcal{L}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\delta}, \nu, \eta; G_h, G_a) &= \mathcal{L}(\boldsymbol{\alpha} - c, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\delta} - c, \nu, \eta + c; G_h, G_a) \\ &= \mathcal{L}(\boldsymbol{\alpha}, \boldsymbol{\beta} - c, \boldsymbol{\gamma} - c, \boldsymbol{\delta}, \nu, \eta + c; G_h, G_a). \end{aligned}$$

To ensure model identifiability, we impose sum-to-zero constraints parametrizing the vectors as:

$$\boldsymbol{\alpha} = \left[\alpha_1, \alpha_2, \dots, \alpha_{n-1}, -\sum_{i=1}^{n-1} \alpha_i \right] \quad (4.16)$$

$$\boldsymbol{\beta} = \left[\beta_1, \beta_2, \dots, \beta_{n-1}, -\sum_{i=1}^{n-1} \beta_i \right] \quad (4.17)$$

$$\boldsymbol{\delta} = \left[\delta_1, \delta_2, \dots, \delta_{n-1}, -\sum_{i=1}^{n-1} \delta_i \right] \quad (4.18)$$

$$\boldsymbol{\gamma} = \left[\gamma_1, \gamma_2, \dots, \gamma_{n-1}, -\sum_{i=1}^{n-1} \gamma_i \right] \quad (4.19)$$

where only the first $n - 1$ parameters are estimated, and the last is determined by the constraint.

In models where η is set equal to zero, the constraints given by Equations 4.17 and 4.18 can be removed, since the restrictions of Equations 4.16 and 4.19 are already sufficient to ensure identifiability. However, in our implementation, we maintain all constraints regardless of whether η is present. This approach ensures consistency across all model variants and enhances interpretability by fixing a reference point (zero mean) for all parameter vectors, allowing parameters to be interpreted as deviations from the average team ability.

4.2.2 Model Variants

Different models arise from simplifying assumptions about the parameters in equations 4.10 and 4.11. The models can be categorized based on three key dimensions:

1. **Offence-Defence Parametrization:** Single parameter ($\boldsymbol{\alpha} = \boldsymbol{\delta}$ and $\boldsymbol{\beta} = \boldsymbol{\gamma}$, with $\boldsymbol{\beta} = -\boldsymbol{\alpha}$) vs. Two parameters ($\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ independent, with $\boldsymbol{\alpha} = \boldsymbol{\delta}$ and $\boldsymbol{\beta} = \boldsymbol{\gamma}$) vs. Four parameters ($\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, $\boldsymbol{\gamma}$, and $\boldsymbol{\delta}$ all independent);
2. **Home Effect:** Present ($\nu \neq 0$) vs. Absent ($\nu = 0$);
3. **Common Effect:** Present ($\eta \neq 0$) vs. Absent ($\eta = 0$).

This section describes all the Poisson model variants (P1–P10) used to model goal scoring in football matches in the present work. These models are organized from the most general formulation, ranging from the simplest single-parameter model, P1, to the most complex four-parameter model with home effect and common effect, P10. The model names were chosen for simplicity, avoiding unnecessary repetition.

Model **P1** uses a single skill parameter $\boldsymbol{\alpha}$ per team, with $\log(\lambda_{i,j}) = \alpha_i - \alpha_j$ and $\log(\lambda_{j,i}) = \alpha_j - \alpha_i = -\log(\lambda_{i,j})$. It assumes identical offensive and defensive abilities, with no home effect or common effect.

Model **P2** extends P1 by adding a home effect parameter ν , giving $\log(\lambda_{i,j}) = \alpha_i + \nu - \alpha_j$ for the home team and $\log(\lambda_{j,i}) = \alpha_j - \alpha_i$ for the away team.

Model **P3** introduces separate offensive (α) and defensive (β) parameters, with $\log(\lambda_{i,j}) = \alpha_i + \beta_j$. It assumes identical abilities at home and away, with no home effect or common effect.

Model **P4** combines P3's two-parameter structure with home effect: $\log(\lambda_{i,j}) = \alpha_i + \beta_j + \nu$ for the home team and $\log(\lambda_{j,i}) = \alpha_j + \beta_i$ for the away team.

Model **P5** uses four independent parameter sets: α, γ for home abilities and β, δ for away abilities, with $\log(\lambda_{i,j,h}) = \alpha_i + \beta_j$ and $\log(\lambda_{j,i,a}) = \gamma_i + \delta_j$. This parametrization already captures home effect implicitly through the difference between home and away parameter sets, making an explicit home effect parameter ν redundant and potentially causing identifiability issues. Model P5 includes no common effect.

Models **P6** through **P10** extend P1 through P5, respectively, by adding a common effect parameter η to both log-intensities: $\log(\lambda_{i,j}) = \log(\lambda_{i,j}^{\text{base}}) + \eta$ and $\log(\lambda_{j,i}) = \log(\lambda_{j,i}^{\text{base}}) + \eta$, where the base structure follows the corresponding model P1–P5. This formulation is inspired by the bivariate Poisson distribution proposed by [Holgate \(1964\)](#), where shared components appear in both rate parameters. In our models, the common effect parameter η acts as a shared additive term on both log-intensities, uniformly scaling both teams' scoring rates by a constant factor $\exp(\eta)$ across all matches. This parameter captures systematic deviations in the overall level of scoring that are not explained by team strengths alone, such as league-wide tactical trends or seasonal effects, rather than match-specific factors such as weather conditions or pitch quality, which would require a match-varying parameter.

The taxonomy of these model variants is illustrated in Figure 7.

4.3 Adaptation to Player-Level Models

The models described in the previous sections operate at the club level, where team abilities are modelled directly as team-specific parameters. However, these models can be adapted to operate at the player level. This adaptation provides a more granular understanding of team performance by decomposing it into individual player contributions.

Instead of estimating team abilities directly, we estimate individual player abilities and aggregate them, weighting by playing time, to obtain team-level strengths. For a given match, let $\mathcal{P}_h$ and $\mathcal{P}_a$ denote the sets of players participating in the home and away teams, respectively, and let m_p denote the number of minutes played by player p in that match. The team-level skill is given by:

$$s_{\text{team}} = \sum_{p \in \mathcal{P}} \exp(\theta_p) \cdot \frac{m_p}{90}, \quad (4.20)$$

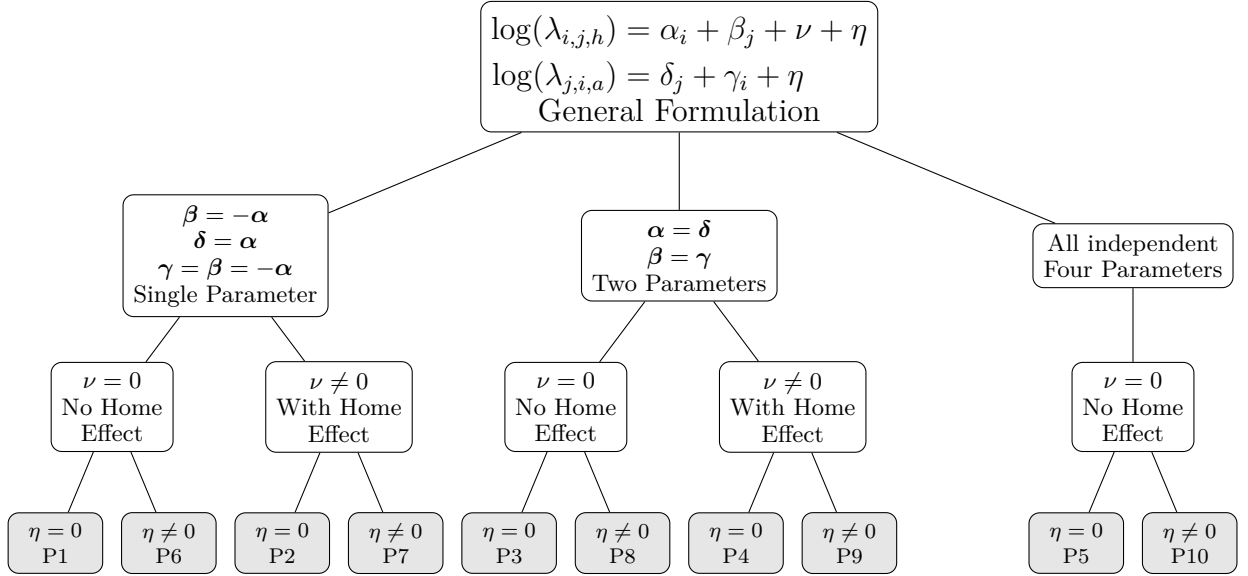


Figure 7 – **Taxonomy of the ten Poisson model variants.** The root node displays the complete four-parameter specification with independent offensive and defensive parameters for home and away contexts (α , β , γ , δ), home effect (ν), and common effect (η). The first level of branching corresponds to the parametrization structure: single parameter (where offensive and defensive abilities are constrained to be equal), two parameters (where home and away abilities are constrained to be equal), or four parameters (where all abilities are independent). The second level distinguishes models with ($\nu \neq 0$) and without ($\nu = 0$) home effect. The third level separates models with ($\eta \neq 0$) and without ($\eta = 0$) common effect. Each leaf node represents a specific model variant, with the model identifier (P1–P10) indicating the final configuration after all simplifications have been applied.

where θ_p is the log-skill parameter for player p , and $\mathcal{P}$ is either $\mathcal{P}_h$ or $\mathcal{P}_a$ depending on the team. The normalization by 90 minutes ensures that a player's contribution is proportional to their playing time relative to the standard match duration.

When a player is sent off (red card), the team plays with fewer players for the remainder of the match. To handle this situation, we introduce a synthetic player with an estimable ability parameter $\theta_{\text{synthetic}}$ that represents the impact of playing with fewer players. Specifically, if a player is sent off at minute t_{off} , we add a synthetic player with $m_{\text{synthetic}} = 90 - t_{\text{off}}$ minutes. If two or more players receive a red card, the same treatment is applied, increasing the value of $m_{\text{synthetic}}$. The synthetic player's ability $\theta_{\text{synthetic}}$ is estimated by the model and quantifies the impact of the red card on team performance. This approach allows the model to learn how much a sending-off affects team strength.

As club-level models, player-level models also require identifiability constraints. The same sum-to-zero constraint is imposed on the player skill parameters:

$$\sum_{p=1}^n \theta_p = 0, \quad (4.21)$$

where n is the total number of players, including the synthetic. This constraint ensures that the model is identifiable and that player abilities are measured relative to the average player ability.

4.4 Inference Procedure

This section describes the prior distributions, data inputs, and computational settings used for posterior inference across all models. All models are fitted within a Bayesian framework using Stan (CARPENTER et al., 2017) via the CmdStanPy interface (STAN DEVELOPMENT TEAM, 2023), with the No-U-Turn Sampler (NUTS) as the Hamiltonian Monte Carlo algorithm.

All club-level models use independent $\text{Normal}(0, 1)$ priors for every parameter. This applies to team log-skill parameters ($\theta, \alpha, \beta, \gamma, \delta$), the home effect (ν), the tie parameter (κ), and the common effect parameter (η), whenever each is present in the model. For player-level Poisson models, the prior on individual skill parameters is set to $\text{Normal}(0, 0.1)$. Since team strength is obtained by aggregating the contributions of 11–16 players (Equation 4.20), the tighter per-player prior ensures numerical stability during posterior sampling. Global parameters (ν and η) retain the $\text{Normal}(0, 1)$ prior at the player level, as their dimensionality does not change. Table 5 summarises the prior distributions used across all models.

Parameter	Description	Club-Level	Player-Level
$\theta, \alpha, \beta, \gamma, \delta$	Skill parameters	$\text{Normal}(0, 1)$	$\text{Normal}(0, 0.1)$
ν	Home effect	$\text{Normal}(0, 1)$	$\text{Normal}(0, 1)$
κ	Tie parameter (BT)	$\text{Normal}(0, 1)$	—
η	Common effect	$\text{Normal}(0, 1)$	$\text{Normal}(0, 1)$

Table 5 – **Prior distributions for all model parameters.** Skill parameters receive a tighter prior at the player level to ensure numerical stability when aggregating individual contributions into team strength. The tie parameter κ is only present in Bradley-Terry models, which are not extended to the player level.

Club-level models receive, for each match, the identifiers of the home and away teams and the number of goals scored by each. For Bradley-Terry models, the goals are collapsed into a categorical outcome (home win, draw, or away win). No information about individual players, substitutions, or playing time is used at this level. Player-level models require a richer data structure. For each match, the model receives the list of players participating in each team, identified by their unique CBF ID, along with the number of minutes played by each player and the total goals scored by each team. This information is derived from the temporal segmentation of match reports described in Section 2.3, which tracks substitutions and red cards to determine each player’s exact playing time. When a

player receives a red card, the synthetic player described in Section 4.3 is assigned the remaining minutes of the match.

The scope of data used for fitting differs between the two analysis contexts. For club-level model evaluation, the rolling-window scheme described in Section 4.5 fits the model incrementally: at each evaluation point, the model receives all *Brasileirão Série A* matches played up to that point in the current season, starting from 50 matches (approximately 5 complete match-days) and progressing through the season in increments of 10 matches. For player-level analysis, the model is fitted on the complete set of matches from the selected period: all 380 matches of the 2025 season for the single-season analysis and all 2,280 matches from 2020 to 2025 for the multi-season analysis. In both cases, only *Brasileirão Série A* data is used.

Club-level models are fitted with 4 parallel chains, each running 1,000 warmup iterations and 1,000 sampling iterations, yielding 4,000 posterior samples per fit. The sampler uses an adaptation target acceptance probability of $\delta = 0.95$ and a maximum tree depth of 10. This configuration is identical to the one used for SBC validation (Section 3.4). Player-level models require a more intensive sampling configuration due to the substantially higher dimensionality of the parameter space. Each fit uses 4 parallel chains with 2,500 warmup iterations and 7,500 sampling iterations per chain, yielding 30,000 posterior samples. The extended warmup allows the sampler to adequately adapt its mass matrix to the high-dimensional geometry, while the larger number of sampling iterations ensures sufficient effective sample size across hundreds of parameters.

4.5 Evaluation Methodology

This section presents the framework used to evaluate the predictive performance of all club-level models on real *Brasileirão Série A* data from 2020 to 2025.

We evaluate model predictions using a rolling-window prediction scheme. After each match-day g , the model is fitted using all matches from rounds 1 to g , and predictions are generated for all future matches. For match-level metrics, we evaluate the predicted probabilities for round $g + 1$; for season-level metrics, the model simulates the remaining $380 - g$ matches to produce predictive distributions of accumulated points for each team at every future match-day. This process begins at round 50 (approximately 5 complete match-days) to ensure sufficient training data, and continues through round 370, yielding predictions for approximately 330 matches per season evaluated out-of-sample. All reported metrics are averaged across evaluation points within each season and then across the five seasons.

Predictions are generated via Monte Carlo integration over the posterior distribution. For each future match, 1,000 parameter vectors are sampled from the posterior. For Bradley-

Terry models, match outcomes are sampled directly from the categorical distribution implied by the model. For Poisson models, home and away goals are simulated independently from Poisson distributions with rates determined by the sampled parameters, and match outcome probabilities are computed as the empirical frequencies of home wins, draws, and away wins. Point totals are accumulated across simulated matches, yielding predictive distributions from which quantile-based intervals are extracted.

To contextualize model performance, we compare against two baseline predictors:

- **Naive 1:** Assigns uniform probabilities $(1/3, 1/3, 1/3)$ to all outcomes, representing a completely uninformative forecast.
- **Naive 2:** Assigns probabilities based on historical outcome frequencies observed in the current season, reflecting the marginal distribution of results up to match-day g .

We assess predictive quality using four complementary scoring rules, all negatively oriented (lower values indicate better predictions). The first three metrics evaluate match-level outcome predictions, while the fourth assesses season-level point forecasts. Let n denote the number of evaluated matches. For each match t , let $\mathbf{z}_t = (z_{H,t}, z_{D,t}, z_{A,t})$ be the one-hot encoded outcome vector, where H , D , and A denote home victory, draw, and away victory, respectively. Let $\mathbf{p}_t = (p_{H,t}, p_{D,t}, p_{A,t})$ be the corresponding predicted probabilities.

The **Brier Score** (BS) measures the accuracy of probabilistic predictions for categorical outcomes, computed as the mean squared error between predicted probabilities and observed outcomes:

$$\text{BS} = \frac{1}{n} \sum_{t=1}^n \sum_{j \in \{H,D,A\}} (z_{j,t} - p_{j,t})^2. \quad (4.22)$$

The **Log Score** (LS), also known as the negative log-likelihood, is a strictly proper scoring rule that heavily penalizes confident but incorrect predictions:

$$\text{LS} = -\frac{1}{n} \sum_{t=1}^n \sum_{j \in \{H,D,A\}} z_{j,t} \log(p_{j,t}). \quad (4.23)$$

The **Ranked Probability Score** (RPS) accounts for the ordinal nature of match outcomes (home win $\succ$ draw $\succ$ away win), penalizing predictions that are “further” from the true outcome:

$$\text{RPS} = \frac{1}{2n} \sum_{t=1}^n \sum_{k=1}^2 \left(\sum_{j=1}^k (z_{j,t} - p_{j,t}) \right)^2. \quad (4.24)$$

Finally, the **Interval Score** (IS) evaluates the quality of predictive intervals for cumulative points. Unlike the previous metrics that assess single-match predictions, IS measures how well the model forecasts the range of possible point totals for each team at future match-days.

A good predictive interval should be both *narrow* (providing precise information) and *calibrated* (containing the true value at the stated confidence level). The IS captures this trade-off: for a confidence level $(1 - \alpha)$, let $[l, u]$ be the predicted interval and y the observed points:

$$\text{IS}_\alpha = (u - l) + \frac{2}{\alpha}(l - y)\mathbf{1}_{y < l} + \frac{2}{\alpha}(y - u)\mathbf{1}_{y > u}. \quad (4.25)$$

The first term penalizes wide intervals, while the second and third terms penalize coverage failures, which are cases where the observed value falls outside the interval. The factor $2/\alpha$ ensures that narrower intervals (smaller α) incur larger penalties when they fail to cover the true value.

We aggregate IS across all 20 teams, all future match-days, and 19 symmetric prediction intervals ranging from 5% to 95% coverage.

For season-level analysis of specific *Brasileirão* seasons, we apply a complementary approach focused on a single well-performing model. After each match-day, the model is refitted using all available match results, and 10,000 Monte Carlo simulations are used to forecast the remaining matches. Each iteration samples parameters from the posterior distribution and simulates all unplayed matches according to the model's likelihood, producing a complete final standings table.

The primary output is a full probability matrix assigning each team a probability for each final position (1–20). From this matrix, we can derive quantities such as:

- Probability of winning the championship;
- Probability of qualifying for *Copa Libertadores da América* (*Libertadores*), South America's premier continental club competition;
- Probability of qualifying for *Copa Sul-Americana* (*Sul-Americana*), South America's secondary continental club competition;
- Probability of relegation.

Additionally, posterior distributions of team strength parameters are obtained after each match-day, providing interpretable summaries of relative team quality.

5 Results

5.1 Simulation-Based Calibration Results

This section presents the results of SBC applied to validate the inference process for all models described in Chapter 4. The SBC procedure was performed independently for each model following the experimental design described in Section 3.4.

5.1.1 Results for Club-Level Models

The SBC validation was performed for all 14 club-level models: 4 Bradley-Terry variants (BT1–BT4) and 10 Poisson variants (P1–P10). As shown in Table 6, no model exhibited more flagged parameters than expected by chance under the 95% confidence bands. Only four models (BT1, P3, P6, and P10) had a single parameter flagged, and in each case, the deviation was minimal: exactly 2 points outside the confidence envelope. These results provide strong evidence that all club-level models are correctly implemented.

Model	Parameters	Expected	Flags	Points Out	Description
BT1	20	1	1	2	Single skill
BT2	21	1	0	0	BT1 + home effect
BT3	21	1	0	0	BT1 + ties
BT4	22	1	0	0	BT1 + home effect + ties
P1	20	1	0	0	Single skill
P2	21	1	0	0	P1 + home effect
P3	40	2	1	2	Attack/defence
P4	41	2	0	0	P3 + home effect
P5	80	4	0	0	Home/away split
P6	21	1	1	2	P1 + common effect
P7	22	1	0	0	P2 + common effect
P8	41	2	0	0	P3 + common effect
P9	42	2	0	0	P4 + common effect
P10	81	4	1	2	P5 + common effect

Table 6 – **SBC validation results for club-level models.** The “Expected” column shows the approximate number of parameters expected to be flagged by chance (5% of total). All models show fewer flags than expected, confirming correct implementation.

To illustrate what a flagged parameter looks like in practice, Figure 8 compares two skill parameters from model P6: α_1 (flagged) and α_2 (not flagged). The visual difference is nearly imperceptible—both curves stay close to zero throughout, with α_1 showing only a

brief excursion beyond the confidence bands. This comparison reinforces that the flagged cases represent statistical noise rather than systematic inference failures.

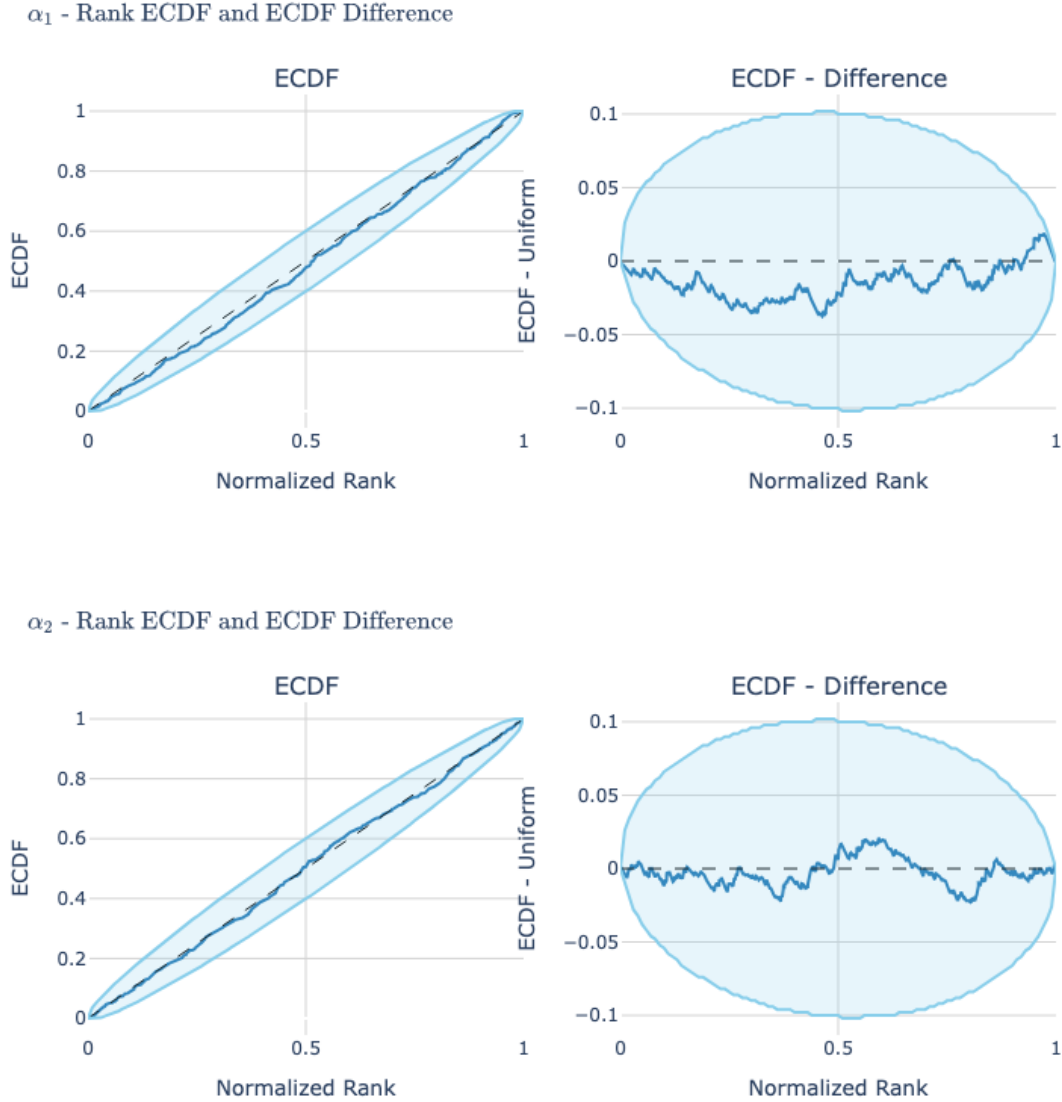


Figure 8 – **ECDF difference plots for model P6**. Top: the skill parameter α_1 , the only flagged parameter in this model, with exactly 2 points marginally outside the envelope. Bottom: the skill parameter α_2 , showing excellent calibration with all ranks within the 95% confidence bands. Left panels show the empirical CDF of normalized ranks; right panels show the difference from the uniform distribution with 95% confidence bands (shaded region). The contrast illustrates that even flagged parameters exhibit only minimal deviations.

5.1.2 Results for Player-Level Models

The player-level models present a significant increase in dimensionality compared to their club-level counterparts. For the SBC validation, synthetic datasets were generated

with 732 unique players participating across the 380 matches of a season. This includes a synthetic player capturing the effect of numerical disadvantage due to red cards.

Table 7 summarizes the SBC results for the 8 player-level Poisson models. Note that we evaluate fewer models at the player level compared to the club level. Due to the high computational cost of player-level inference (hundreds to thousands of parameters per model), we restricted validation to a subset of models based on the club-level evaluation on real data (Section 5.2). Bradley-Terry models were excluded as they performed below the Naive 2 baseline on BS, RPS and LS, indicating poor predictive ability. Models P5 and P10 were also excluded due to severe over-fitting caused by their home/away split parametrization, performing worse than an uninformative baseline. The remaining 8 Poisson models (P1–P4, P6–P9) were retained: models with home effect (P2, P4, P7, P9) for their strong predictive performance, and models without home effect (P1, P3, P6, P8) to validate the base model structures.

Despite the substantially higher number of parameters—ranging from 732 to 1,466 per model—the flagging rates remain low and consistent with chance. For instance, model P9 has 1,466 parameters and would be expected to have approximately 73 flagged parameters (5%) under the null hypothesis; only 10 were observed. Similarly, model P3 with 1,464 parameters showed only 6 flags, far below the expected 73.

Model	Parameters	Expected	Flags	Points Out	Description
P1	732	37	3	6	Single skill
P2	733	37	5	10	P1 + home effect
P3	1,464	73	6	14	Attack/defence
P4	1,465	73	8	16	P3 + home effect
P6	733	37	4	8	P1 + common effect
P7	734	37	4	8	P2 + common effect
P8	1,465	73	8	16	P3 + common effect
P9	1,466	73	10	20	P4 + common effect

Table 7 – **SBC validation results for player-level models.** Despite the high dimensionality (up to 1,466 parameters), all models show far fewer flags than the expected 5% rate.

The flagged parameters correspond to individual player skills (α_i or β_i) distributed across different simulated players, with no systematic pattern. Also, each flagged parameter had minimal deviations, as observed in club-level models. These results confirm that the extension to high-dimensional player-level inference does not compromise calibration.

5.2 Model Evaluation on Real Data

After validating the inference procedure through Simulation-Based Calibration in Section 5.1, we now evaluate the predictive performance of all club-level models on real

Brasileirão Série A data from 2020 to 2025 using the evaluation methodology described in Section 4.5.

Models BT1 and BT2 are important from a theoretical standpoint as the simplest Bradley-Terry specifications. Their successful SBC validation (Section 5.1.1) confirms the correctness of the inference framework before extending to more complex variants. However, for the real data evaluation, these models were excluded because they do not allow ties, a result that occurs with non-negligible frequency in football. Table 8 presents the predictive performance of all 12 club-level models and the two baselines.

Model	BS	RPS	LS	IS
Naive 1	0.6667	0.2359	1.0986	1086.2815
Naive 2	0.6373	0.2243	1.0570	1079.1856
BT3	0.6494	0.2293	1.0768	1073.2671
BT4	0.6508	0.2296	1.0800	1075.1171
P1	0.6497	0.2287	1.0741	1061.8859
P2	0.6256	0.2180	1.0402	1059.8783
P3	0.6558	0.2324	1.0860	1082.7855
P4	0.6356	0.2230	1.0550	1078.7061
P5	0.6834	0.2460	1.1255	1103.8898
P6	0.6495	0.2286	1.0738	1061.5644
P7	0.6237	0.2172	1.0378	1061.0630
P8	0.6548	0.2318	1.0840	1081.7105
P9	0.6334	0.2218	1.0526	1084.7895
P10	0.6835	0.2460	1.1263	1104.6147

Table 8 – **Predictive performance of club-level models on *Brasileirão* data (2020–2025)**. Models are grouped by type: naive baselines, Bradley-Terry (BT), and Poisson (P). Lower values indicate better performance; best values for each metric are shown in bold.

These results guide the selection of models for player-level extension. First, models P5 and P10, which use separate skill parameters for home and away contexts instead of an explicit home effect parameter, perform poorly across all metrics, ranking below even the uninformative baseline (Naive 1). This suggests that the additional flexibility leads to overfitting given the limited number of matches per team in a single season. Second, Bradley-Terry models consistently underperform compared to the best Poisson models, likely because they discard valuable information by modelling only match outcomes rather than goal counts. Third, among Poisson models, those without an explicit home effect parameter (P1, P3, P6, P8) are consistently outperformed by their counterparts with home effect (P2, P4, P7, P9). This confirms that explicitly modelling home effect is essential for predictive accuracy in the *Brasileirão*.

Based on these findings, only Poisson models with home effect were selected for further evaluation on real data: P2, P4, P7, and P9. These four models span the two main

parametrization structures, single skill (P2, P7) and attack/defence (P4, P9), and both common effect settings, without common effect (P2, P4) and with common effect (P7, P9). Also, for more detailed analysis on club level, we will use only the model P7, the best model according to Table 8.

5.3 Match-Level Parameters

Beyond team-specific strength parameters, the P7 model includes two match-level parameters that capture important structural features of football: home effect (ν) and a common effect parameter on both teams' goal rates (η). This section analyses the posterior distributions of these parameters across the 2020–2025 *Brasileirão* seasons.

5.3.1 Home Effect

The home effect is a well-documented phenomenon in football, where teams playing at their own stadium tend to perform better due to factors such as crowd support, familiarity with the pitch, and reduced travel fatigue (POLLARD, 1986). In the P7 model, this effect is captured by the parameter ν , which adds a constant bonus to the log-expected goals of the home team. The multiplicative effect on expected goals is given by $\exp(\nu)$. For example, a value of $\nu = 0.30$ implies that the home team is expected to score $\exp(0.30) \approx 1.35$ times more goals than if playing away, holding all other factors constant, i.e., playing against the same team but as an away team.

Figure 9 shows the posterior distributions of ν across the analysed seasons. The 2020 season, played without spectators due to the COVID-19 pandemic from August 2020 through February 2021, exhibits a median $\nu = 0.273$ ($\exp(\nu) \approx 1.31$). Studies of European leagues during the same period reported substantial reductions in home advantage during closed-door matches (FISCHER; HAUCAP, 2021; SORS et al., 2021; BENZ; LOPEZ, 2023). However, this effect is not clearly observed in the *Brasileirão* data: the 2020 estimate is similar to 2023 ($\nu = 0.281$), a season played under normal conditions, and the posterior distributions overlap considerably. This suggests that crowd presence may not be the dominant driver of home advantage in Brazilian football. Alternatively, factors such as differences in stadium infrastructure, travel distances, or tactical adaptations, may have compensated for the absence of supporters.

Table 9 summarises the posterior statistics for both match-level parameters across all seasons.

5.3.2 Common Effect on Goals

The common effect parameter η captures deviations in total goals scored from what would be expected based on team strengths alone. In the P7 model, η affects both teams'

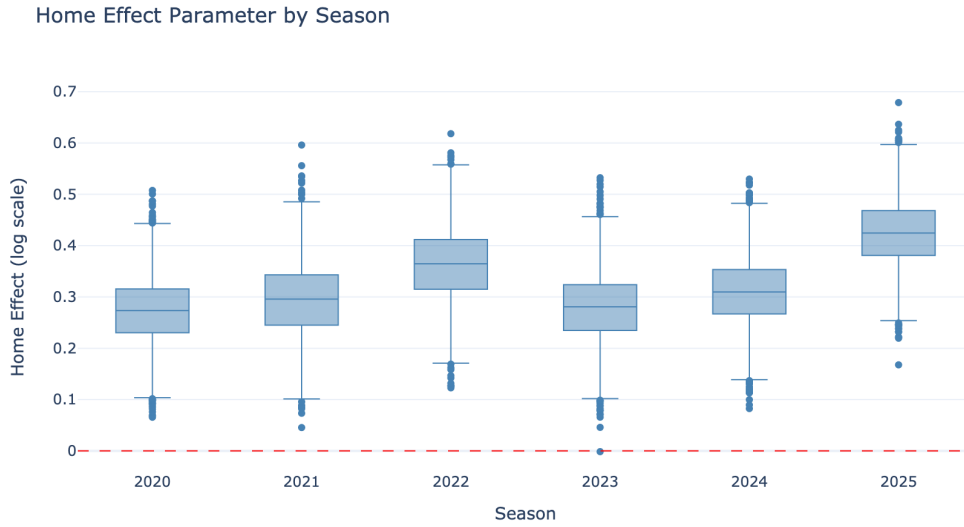


Figure 9 – **Posterior distributions of the home effect parameter (ν) across seasons.** Each boxplot shows the posterior distribution estimated from 4,000 MCMC samples. The dashed horizontal line at zero represents no home effect. Higher values indicate stronger home advantage.

expected goals symmetrically. Positive values indicate that matches tend to produce more goals, while negative values suggest fewer goals than predicted by team quality differences.

Figure 10 reveals that η fluctuates around zero across seasons, with both positive and negative values observed. The 2020 and 2023 seasons exhibit slightly positive median values ($\eta \approx 0.03$), corresponding to $\exp(\eta) \approx 1.03$, i.e., approximately 3% more goals than expected. In contrast, 2021 shows the most negative value (median $\eta = -0.096$, corresponding to $\exp(\eta) \approx 0.91$), i.e., roughly 9% fewer goals than predicted by team strengths.

These variations may reflect changes in tactical approaches, referee behaviour, or overall league competitiveness. A negative η could indicate a more defensive league style or greater parity among teams, leading to tighter, lower-scoring matches. The near-zero values observed in most seasons suggest that the team strength parameters already capture the primary drivers of goal-scoring, with η making only modest adjustments. This hypothesis is supported by the fact that the results of the models with common effect are very similar to the results of their counterparts without common effect.

5.4 Club-Level Applications

This section demonstrates the practical application of club-level models to real *Brasileirão* seasons from 2020 to 2025. Using one of the best-performing models from Section 5.2 (P7) and the season-level simulation approach described in Section 4.5, we

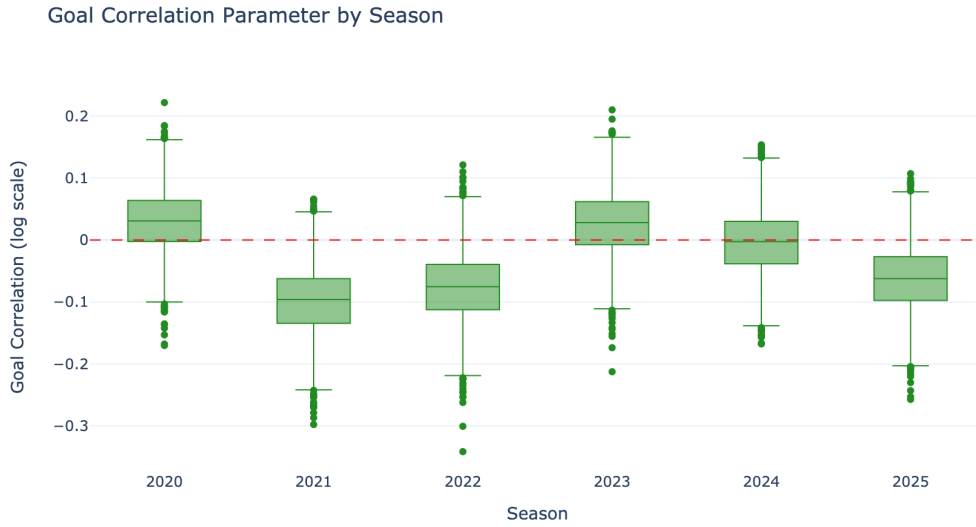


Figure 10 – **Posterior distributions of the common effect parameter (η) across seasons.** Each boxplot shows the posterior distribution estimated from 4,000 MCMC samples. The dashed horizontal line at zero represents no systematic deviation from expected goals. Positive values indicate higher-scoring matches; negative values indicate lower-scoring matches.

Season	ν	$\exp(\nu)$	η	$\exp(\eta)$
2020	0.273 (0.147, 0.402)	1.31	0.031 (-0.069, 0.127)	1.03
2021	0.296 (0.162, 0.430)	1.34	-0.096 (-0.204, 0.004)	0.91
2022	0.365 (0.226, 0.497)	1.44	-0.075 (-0.182, 0.035)	0.93
2023	0.281 (0.142, 0.404)	1.32	0.028 (-0.071, 0.126)	1.03
2024	0.310 (0.185, 0.441)	1.36	-0.003 (-0.105, 0.091)	1.0
2025	0.425 (0.293, 0.548)	1.53	-0.062 (-0.162, 0.039)	0.94

Table 9 – **Summary statistics for match-level parameters by season.** Values shown are posterior medians with 2.5th and 97.5th percentiles in parentheses.

illustrate how the model generates probabilistic forecasts for championship outcomes. While Section 5.2 evaluated all club-level models across multiple seasons using quantitative scoring rules, here we focus on a single well-performing model and examine specific seasons in depth, providing qualitative insights into how probabilistic forecasts evolve and how the model captures the real championship race. We select specific seasons that showcase different aspects of the model’s behaviour: competitive title races, historic relegations, and the inherent uncertainty of football. Other seasons have their graphics available in the Appendix A.

5.4.1 2020 Season

The 2020 season was played under the unique circumstances of the COVID-19 pandemic, with a calendar that started in August 2020 and extended into February 2021.

This season produced one of the most competitive title races in recent memory, with four clubs trading positions as favourites throughout the campaign.

Figure 11 shows how Internacional, São Paulo, Atlético Mineiro, and Flamengo alternated as title favourites. No club maintained a sustained lead: Atlético Mineiro reached approximately 70% probability in October, while São Paulo and Internacional reached near 80% in December and January, respectively. Flamengo, who had remained competitive throughout the season, secured the title in the final rounds. The volatility of these probability trajectories, with the leading club changing multiple times, reflects the genuine competitive balance among the top teams.

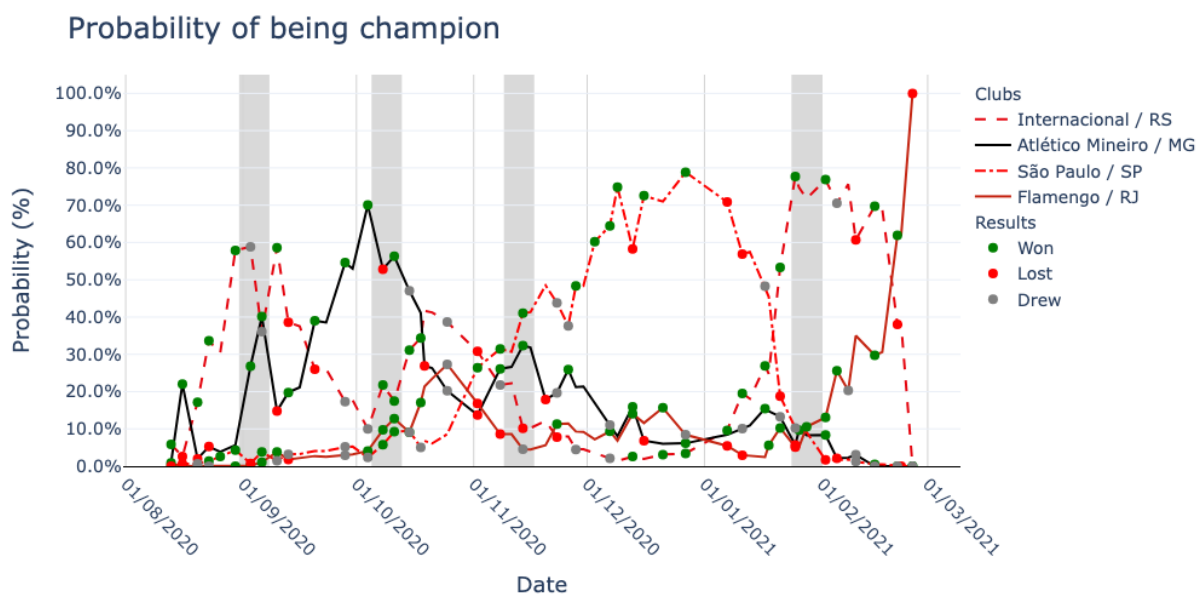


Figure 11 – **Championship probability evolution for the 2020 *Brasileirão***. Each line represents the probability of winning the title for a given club, updated after each match-day. Markers indicate individual match results: green circles denote victories, red circles denote defeats, and gray circles denote draws. Shaded vertical bands indicate FIFA international periods.

The relegation battle exhibited distinct probability trajectories for different clubs, as shown in Figure 12. The high probabilities observed in August (Fortaleza over 80%, Sport and Goiás near 70%) reflect estimation uncertainty due to limited data: at this stage, each team had played only a few matches, and the posterior estimates are dominated by prior information rather than observed performance. As more matches were played, the estimates converged to more stable values. Goiás exhibited a steady increase in relegation probability from October onwards, reaching near-certainty by the final rounds and being mathematically relegated before the last match-day. Sport maintained high probabilities during the second half of the season. However, Sport's results improved in the closing stages of the season, while Vasco da Gama's form deteriorated, resulting in Sport avoiding relegation. On the other hand, Vasco da Gama's trajectory illustrates a progressive deterioration: starting with near-zero probability in August, their estimated

relegation risk increased throughout the season, culminating in their relegation. Fortaleza's probability stabilized at low values once sufficient data accumulated, and they finished comfortably above the relegation zone despite losing their final three matches.

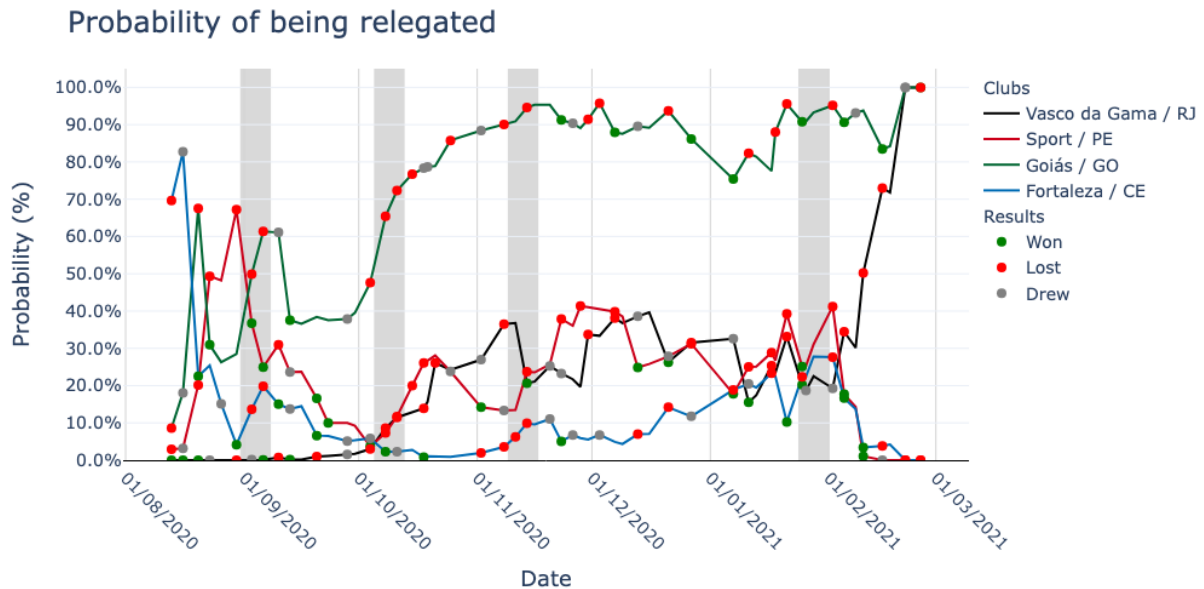


Figure 12 – **Relegation probability evolution for the 2020 *Brasileirão***. Each line represents the probability of relegation (finishing in positions 17–20) for a given club, updated after each match-day. Markers indicate individual match results: green circles denote victories, red circles denote defeats, and gray circles denote draws. Shaded vertical bands indicate FIFA international break periods.

Figure 13 shows the mid-season position distribution for all 20 clubs after 192 matches (approximately half the season). Each cell represents the probability of a club finishing in a given position, with brighter colours indicating higher probabilities.

The heatmap reveals a clear stratification of the league at mid-season. At the top, São Paulo leads with 31.46% probability of winning the title, followed by Atlético Mineiro (26.11%), Internacional (21.78%), and Flamengo (8.61%). São Paulo shows the highest concentration in the top four positions (79.31%), followed by Atlético Mineiro (78.23%) and Internacional (74.83%). Flamengo, while still a title contender, exhibits a more diffuse distribution with only 52.96% probability of finishing in the top four, extending into positions 5–7. Palmeiras shows probability mass concentrated in positions 4–8, with 59.84% probability of finishing in the top six, indicating a strong campaign at mid-season.

The middle of the table displays considerable uncertainty, though with small identifiable clusters. Fluminense and Grêmio show their probability mass concentrated in positions 4–8, while Santos exhibits a similar concentration shifted slightly lower to positions 5–9. Further down, Fortaleza's distribution places its mass of probability in

positions 7–11. These wide intervals, spanning 5 to 7 positions each, reflect the difficulty in distinguishing teams of similar quality based on partial-season data.

At the bottom, the relegation picture shows varying degrees of certainty. Goiás exhibits highly concentrated probability mass in position 20 and has 90.02% probability of relegation (positions 17–20). Botafogo (42.81% relegation probability), Red Bull Bragantino (40.94%), Coritiba (41.54%), and Atlético Paranaense (51.53%) show distributions concentrated in the lower positions, indicating significant relegation risk.

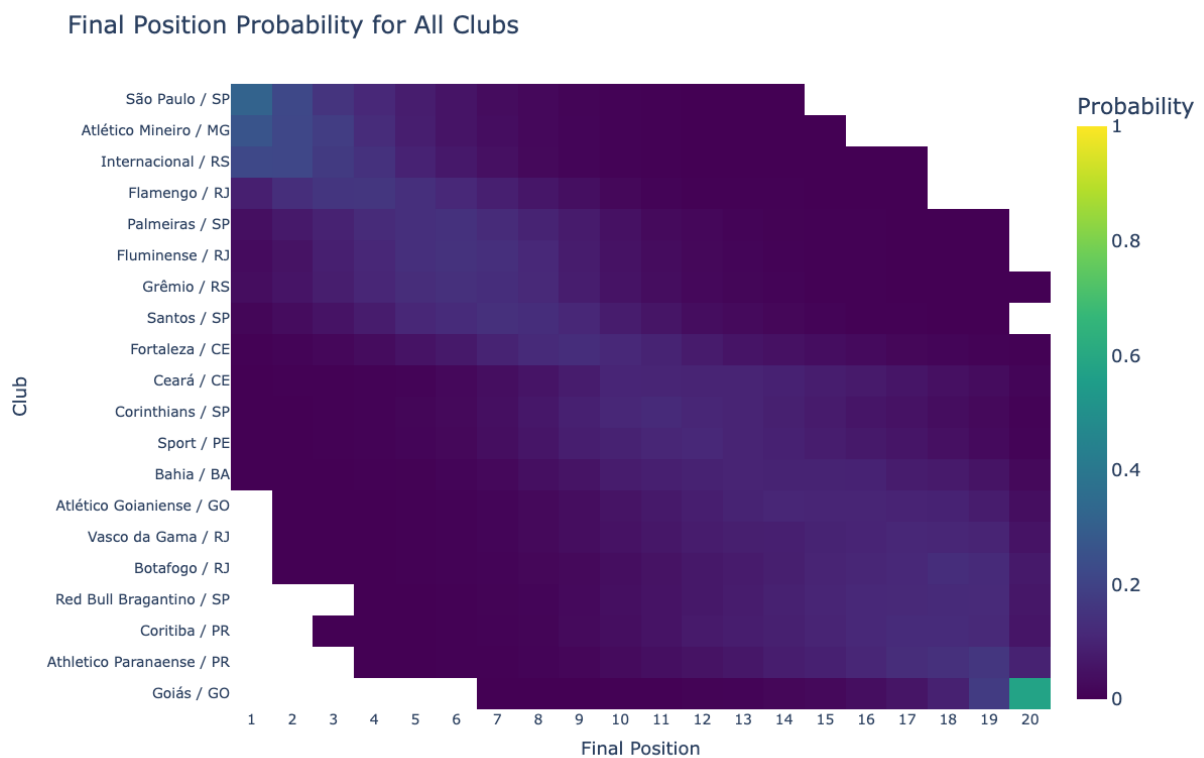


Figure 13 – **Final position probability distribution at mid-season 2020 (after 192 matches)**. Each cell shows the probability of a club (row) finishing in a given position (column), estimated from 10,000 Monte Carlo simulations. Clubs are ordered vertically by expected final position, with brighter colours indicating higher probabilities.

The end-of-season team strength estimates are shown in Figure 14. The box-plots display posterior distributions of the team strength parameter (in log scale) for each club, ordered by estimated median strength.

The figure reveals the competitive nature of this season through the extensive overlap of posterior distributions. Among the title contenders, Internacional shows the highest median strength, but the median strengths of Flamengo up to Red Bull Bragantino are all very similar, indicating that these clubs possessed similar underlying quality.

In the middle of the table, something similar happens to Ceará, Atlético Paranaense, Santos and Corinthians. Also, Atlético Goianiense, Fortaleza and Bahia show similar me-

dian strengths. At the bottom, the relegated clubs (Vasco da Gama, Goiás, Coritiba and Botafogo) join Sport to form another group of clubs with similar median strengths.

The whisker ranges of Internacional and Botafogo do not overlap with each other, nor with most other clubs: Internacional's distribution lies entirely above the average (zero), while Botafogo's lies entirely below. For the remaining clubs, considerable overlap exists among the whisker ranges, particularly within the mid-table group, making it difficult to distinguish teams based on strength estimates alone.

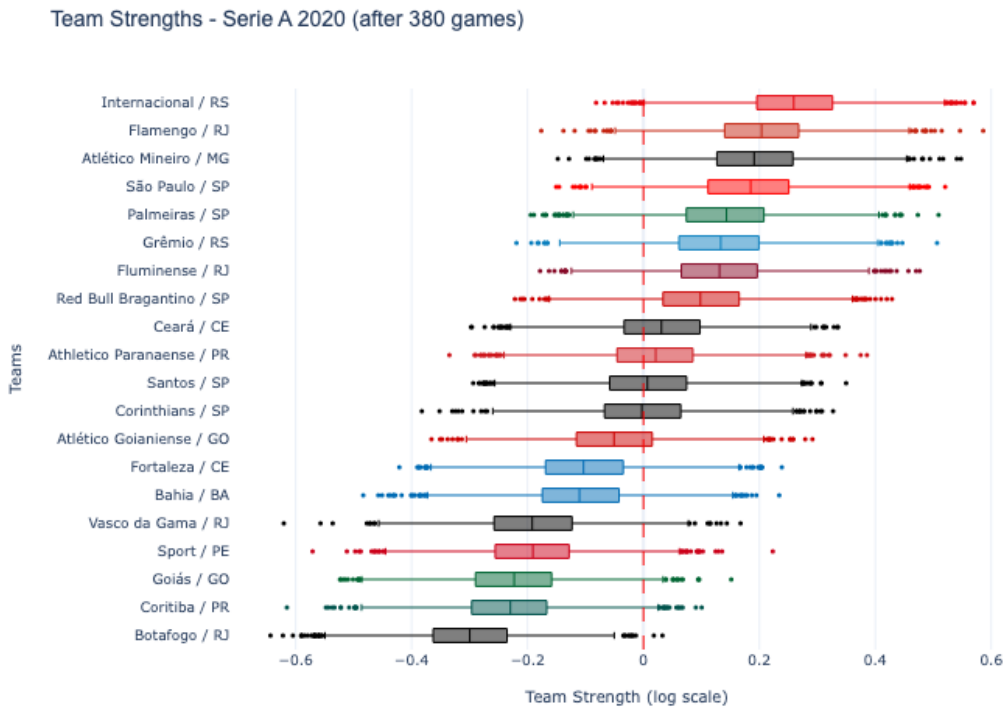


Figure 14 – **Posterior distributions of team strengths for the 2020 *Brasileirão* (after 380 matches)**. Each box-plot shows the posterior distribution of the team strength parameter (log scale) estimated from 4,000 MCMC samples. Teams are ordered vertically by posterior median strength, from lowest (bottom) to highest (top). The box represents the interquartile range (IQR), the vertical line inside the box indicates the median, and whiskers extend to 1.5 times the IQR. The dashed red vertical line at zero represents the average team strength.

5.4.2 2023 Season

While the 2020 season had a competitive title race, the seasons 2021 and 2022 saw two solid campaigns, with titles for Atlético Mineiro and Palmeiras, respectively. The season of 2023 was on track to be another solid campaign, with the high performance of Botafogo, with a difference of 13 points to the second place after 19 matches. However, the season produced one of the most significant late-season reversals in Brazilian football history.

Figure 15 shows the championship probability evolution, dominated by Botafogo from June through October. By mid-June, Botafogo had a probability of over 60% to win the title. With three consecutive victories, Botafogo's probability rose to 90% in mid-July. During August, Botafogo's probability remained above 90%, exceeding 95% in some points. September proved challenging, with three consecutive defeats and a draw, but the probability remained around 80%. In October, with two consecutive victories, Botafogo's probability returned to above the 90%, but from October 21 on, Botafogo stopped winning. In the season's last 11 matches, Botafogo drew six and lost five, including a sequence of four consecutive defeats. During this period, the second place team, Palmeiras, was able to close the gap by winning 8 matches, including a 3–4 victory over Botafogo, having trailed 3-0 at half-time. At the end of the season, Palmeiras was the champion, and Botafogo finished in fifth place.

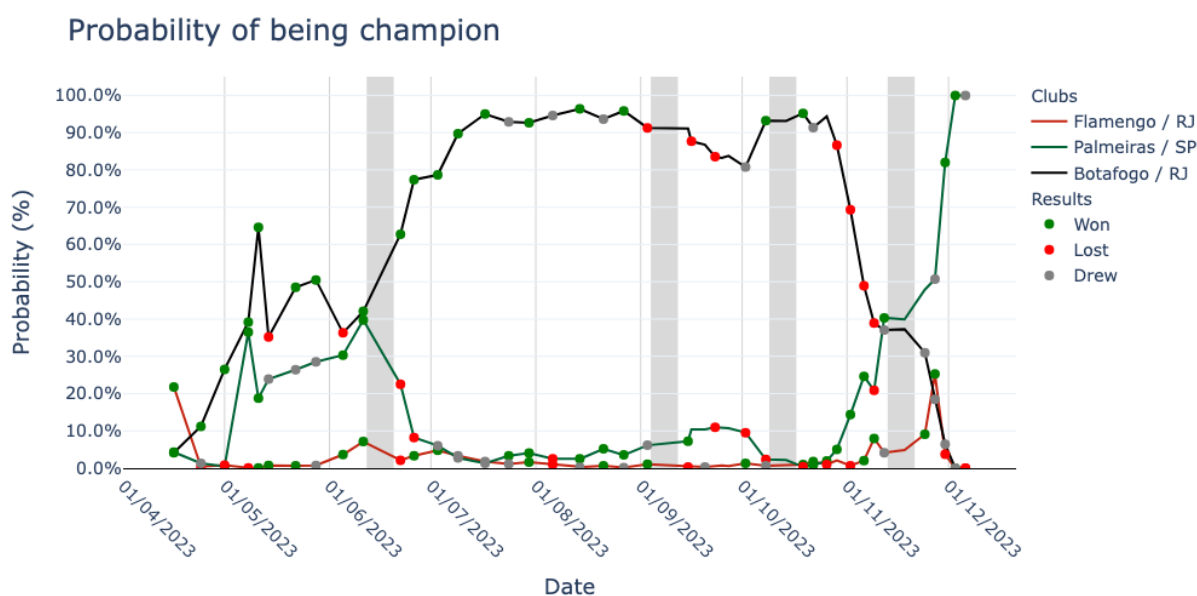


Figure 15 – **Championship probability evolution for the 2023 *Brasileirão*.** Each line represents the probability of winning the title for a given club, updated after each match-day. Markers indicate individual match results: green circles denote victories, red circles denote defeats, and gray circles denote draws. Shaded vertical bands indicate FIFA international break periods.

The relegation battle was equally dramatic, culminating in the first relegation in Santos's 111-year history. Figure 16 shows how the relegation probabilities of Vasco, Bahia, Goiás and Santos oscillated throughout the season. With a collapse in the title race, the relegation battle also produced an unexpected outcome. Remaining only one round, América-MG, Coritiba and Goiás were already relegated. The last club to be relegated was among Bahia, Vasco and Santos. Bahia were in the relegation zone, with a probability of 59.60% of being relegated. Vasco, the first club out of the relegation zone, had a probability of 28.91%. Finally, Santos were the favourite to avoid relegation, with a probability of 11.49%.

On the final round, Santos could avoid relegation by winning their match or, if drawing their match, by Vasco or Bahia not winning their matches. If Santos lost, they still could avoid relegation if Vasco lost or with Bahia not winning their match. At the final round, Bahia won the match against Atlético Mineiro, while Vasco won against Red Bull Bragantino. With a loss to Fortaleza, Santos were relegated.

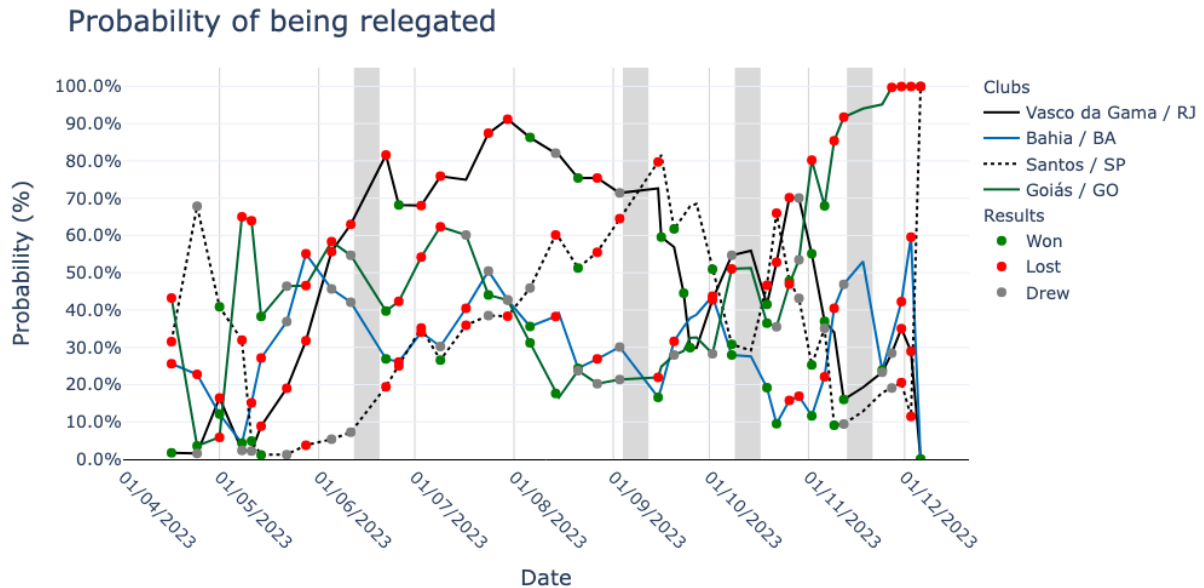


Figure 16 – **Relegation probability evolution for the 2023 *Brasileirão***. Each line represents the probability of relegation (finishing in positions 17–20) for a given club, updated after each match-day. Markers indicate individual match results: green circles denote victories, red circles denote defeats, and gray circles denote draws. Shaded vertical bands indicate FIFA international break periods.

Figure 17 shows the mid-season position distribution for all 20 clubs after 188 matches. The contrast with 2020 is striking: while 2020 exhibited diffuse probabilities at the top reflecting a four-way race for the title, the 2023 heatmap shows extreme concentration, especially on the first and second places. Botafogo holds 95.98% probability of winning the title, with the remaining probability distributed between Palmeiras (2.83%) and negligible amounts for other clubs. Palmeiras shows probability mass concentrated in positions 2–4, with 79.46% probability of finishing in the top four.

At the bottom, the relegation picture shows América-MG with near-certain probability of finishing in the relegation zone (96.73% in positions 17–20). Coritiba, with 88.17%, and Vasco, with 81.67%, also show high relegation probabilities. Santos exhibits 59.44% relegation probability at mid-season. Notably, Bahia shows 39.16% relegation probability at that point, while Goiás had 16.51% probability of being relegated. These values increased at the end of the season due to some losses.

The end-of-season team strength estimates are shown in Figure 18. The box-plots display posterior distributions of the team strength parameter (in log scale) for each club,

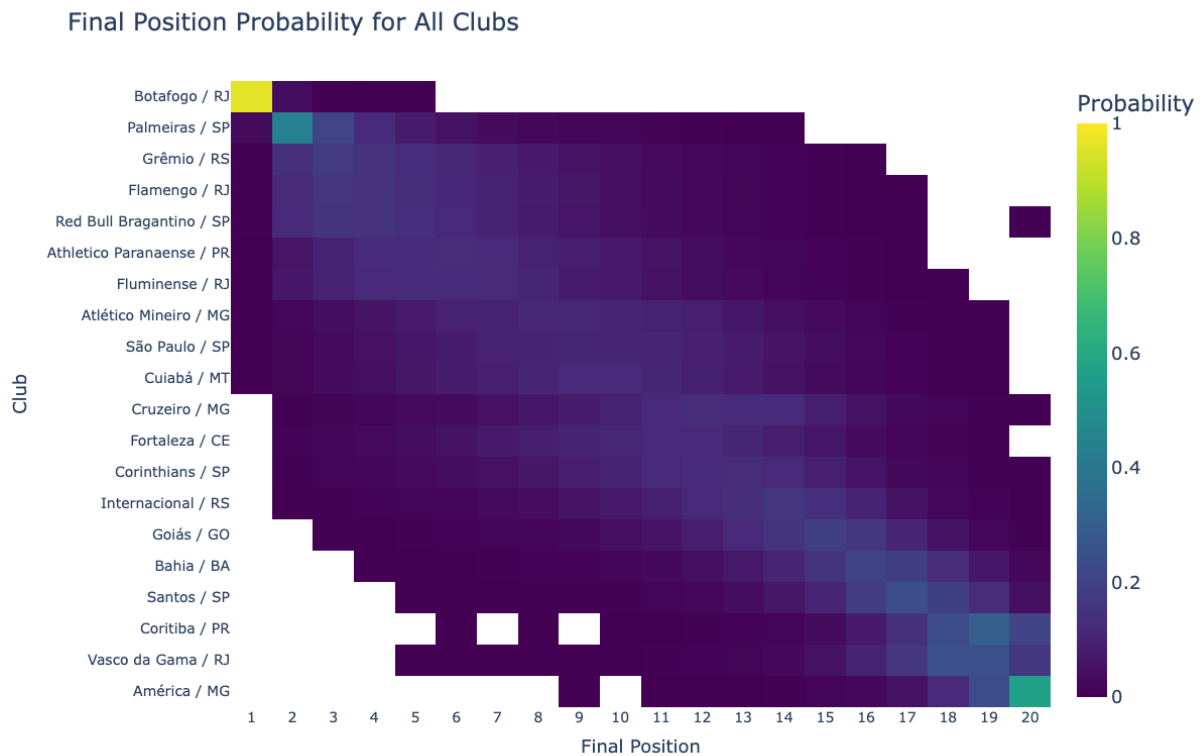


Figure 17 – **Final position probability distribution at mid-season 2023 (after 188 matches)**. Each cell represents the estimated probability of a club (row) finishing the season in a given position (column), based on 10,000 Monte Carlo simulations of the remaining 192 matches. Clubs are ordered vertically by expected final position; brighter colours indicate higher probabilities.

ordered by posterior median strength.

Like 2020, 2023 does not show a clear separation at the top. Palmeiras exhibits the highest estimated median strength, followed by Botafogo and Atlético Mineiro, whose distributions show considerable mutual overlap. Flamengo and Red Bull Bragantino form another tier, with their whisker ranges overlapping substantially with the top three but median strengths lower. In the middle of the table, São Paulo, Internacional, Fortaleza, Cuiabá and Corinthians are around the average (zero), with extensive overlap making it difficult to distinguish between these clubs based on strength estimates alone.

Vasco da Gama occupies the bottom positions among the non-relegated clubs, with their distribution showing some overlap with the relegated teams. At the bottom, the four relegated clubs (Goiás, Santos, Coritiba, and América-MG) show lower strength estimates. América-MG's distribution lies entirely below the average (zero), while Coritiba's has some outliers above the average. However, both clubs have a clear separation from Palmeiras.

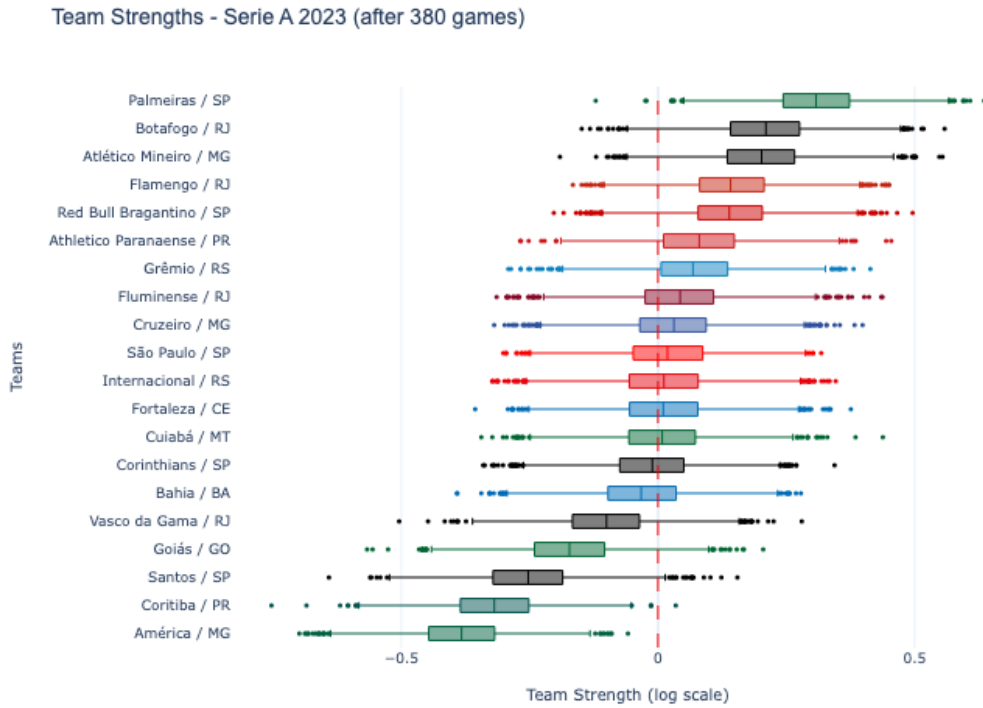


Figure 18 – **Posterior distributions of team strengths for the 2023 *Brasileirão* (after 380 matches)**. Each boxplot shows the posterior distribution of the team strength parameter (log scale) estimated from 4,000 MCMC samples. Teams are ordered vertically by posterior mean strength, from lowest (bottom) to highest (top). The box represents the interquartile range (IQR), the vertical line inside the box indicates the median, and whiskers extend to 1.5 times the IQR. The dashed red vertical line at zero represents the average strength.

5.4.3 2025 Season

If Botafogo collapsed in 2023, in 2024 they had a season with intense competition for the title with Palmeiras. They won the championship on the last round, after a victory over São Paulo. The year of Botafogo was also marked by the title of Copa Libertadores, with a 3-1 victory over Atlético Mineiro in the final. After an eventful year in 2024, we select 2025 as a season with similar title race and because it is more recent.

The 2025 season featured an intense battle between Flamengo and Palmeiras for the title. In addition to the Brasileirão, Flamengo and Palmeiras also contested the final of Copa Libertadores, with Flamengo winning the title by 1-0. For Brazilian clubs, these two titles are the most important ones of the season, showing the strength of these clubs.

Figure 19 shows the championship probability evolution. In July, after the FIFA Club World Cup, Flamengo had almost 80% probability of winning the title. At this moment, Flamengo had 27 points in 12 games, while Cruzeiro had 24 points in 12 games and Palmeiras 22 points in 11 games. Until October, Flamengo was leading the championship with similar performance of Palmeiras. On October 1, Flamengo had 54 points and

Palmeiras had 52 points, both with 24 games played. At this point, both clubs had a similar probability of winning the title.

This equilibrium followed until mid-November, with a jump of Flamengo's probability in mid-October, after a 3-2 victory over Palmeiras, followed by a fall after a loss by 1-0 to Fortaleza. In November, after a sequence of five consecutive matches without victory, Palmeiras' probability of being champion fell to near zero. At the penultimate round, Flamengo guaranteed the title by winning their match against Ceará by 1-0.

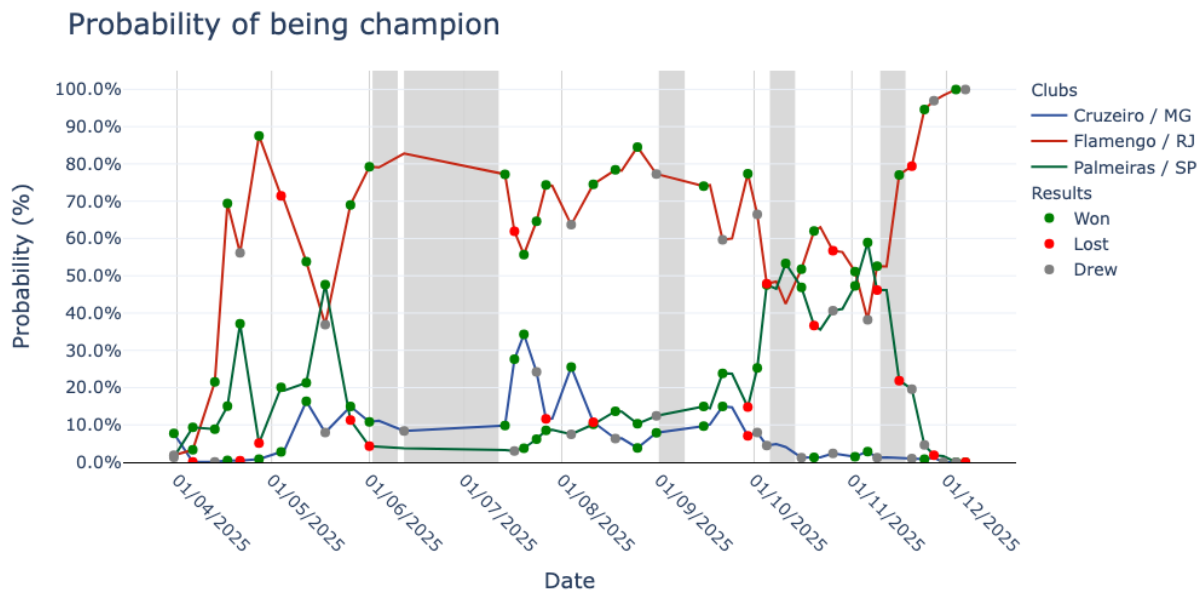


Figure 19 – **Championship probability evolution for the 2025 *Brasileirão***. Each line represents the probability of winning the title for a given club, updated after each match-day. Markers indicate individual match results: green circles denote victories, red circles denote defeats, and gray circles denote draws. Shaded vertical bands indicate FIFA international break periods.

Like 2023, the relegation battle only finished in the last round. In this season, the battle was for two positions on the relegation zone, with six clubs with mathematical chances of being relegated: Internacional, Vitória, Fortaleza, Ceará, Santos and Atlético Mineiro. Atlético's probability was near zero, since they had 3 points more than Vitória, first club in the relegation zone, and a goal difference of -6 goals, while Vitória had a goal difference of -18 goals. Santos also had a small chance, with a probability of 1.78%. Ceará and Fortaleza, both with only one point over the relegation zone, had a probability of 15.71% and 50.92%, respectively. The two clubs that were in the relegation zone, Vitória and Internacional, had a probability of 58.27% and 73.32%, respectively. In the last round, both Vitória and Internacional won their matches, while Fortaleza and Ceará lost, resulting in the relegation of the two rivals.

Figure 21 shows the mid-season position distribution for all 20 clubs after 189 matches. Due to the FIFA Club World Cup, some matches were rearranged along the season. Then at this point, Flamengo had 43 points in 19 games, while Palmeiras had 39

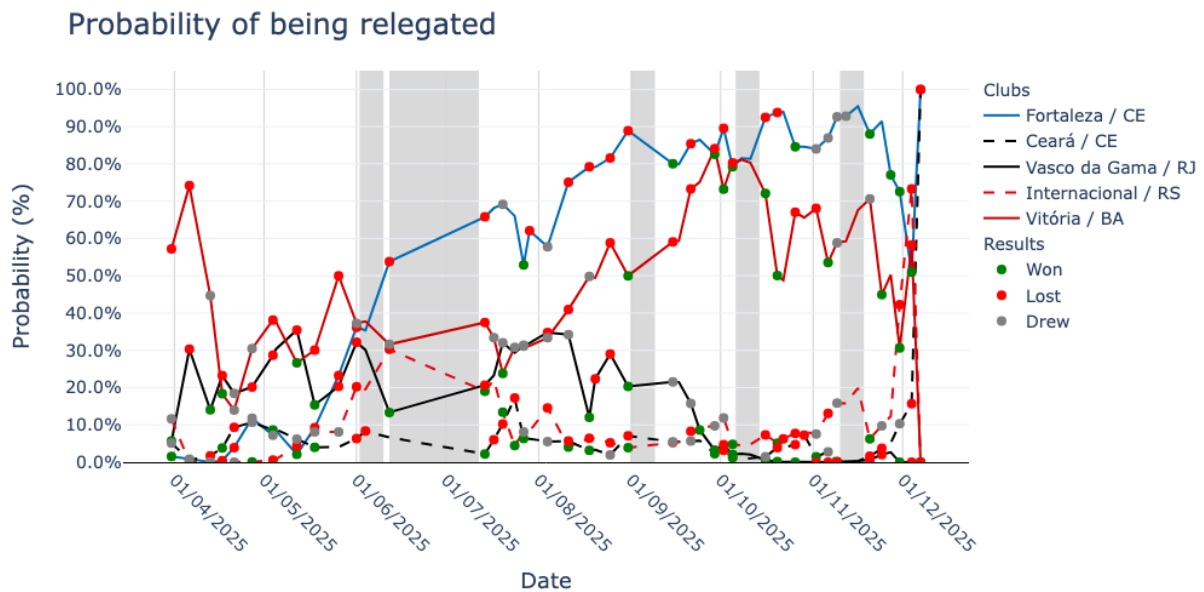


Figure 20 – **Relegation probability evolution for the 2025 *Brasileirão***. Each line represents the probability of relegation (finishing in positions 17–20) for a given club, updated after each match-day. Markers indicate individual match results: green circles denote victories, red circles denote defeats, and gray circles denote draws. Shaded vertical bands indicate FIFA international break periods.

in 18 matches and Cruzeiro had 38 in 20 matches. Since Flamengo's points are already guaranteed, the probability of Flamengo winning the title, 78.19%, has a slight bias when compared to Palmeiras, 13.61%, whose points are yet to be confirmed or Cruzeiro, 6.38%, which had already played one more game. Palmeiras and Cruzeiro are the favourites to finish in second and third place, respectively.

A notable feature of 2025 is Mirassol's strong performance. The club, promoted from *Série B*, shows probability concentrated in positions 4–6, representing one of the best campaigns for a debuting team. Bahia and Botafogo also occupy positions 4–6, but with a slightly higher probability of finishing in the top four.

At the bottom, Sport shows near-certain relegation probability (95.95% in positions 17–20). Juventude, 79.06%, and Fortaleza, 79.13%, also exhibit high relegation probabilities. Vitória shows 49.45% relegation probability, reflecting their precarious but not hopeless position. Santos, 25.12%, Vasco, 22.38% and Corinthians, 20.36%, appear in danger but with substantial probability of survival. At this point, Internacional had only 6.33% probability of being relegated.

The end-of-season team strength estimates are shown in Figure 22. Flamengo and Palmeiras exhibit the highest estimated strengths, with distributions clearly above the average (zero) and clearly separated from Juventude and Sport. Flamengo also can be distinguished from Red Bull Bragantino, Internacional, Fortaleza and Vitória. Cruzeiro, Mirassol and Botafogo form a second group of clubs with similar strengths, all with a

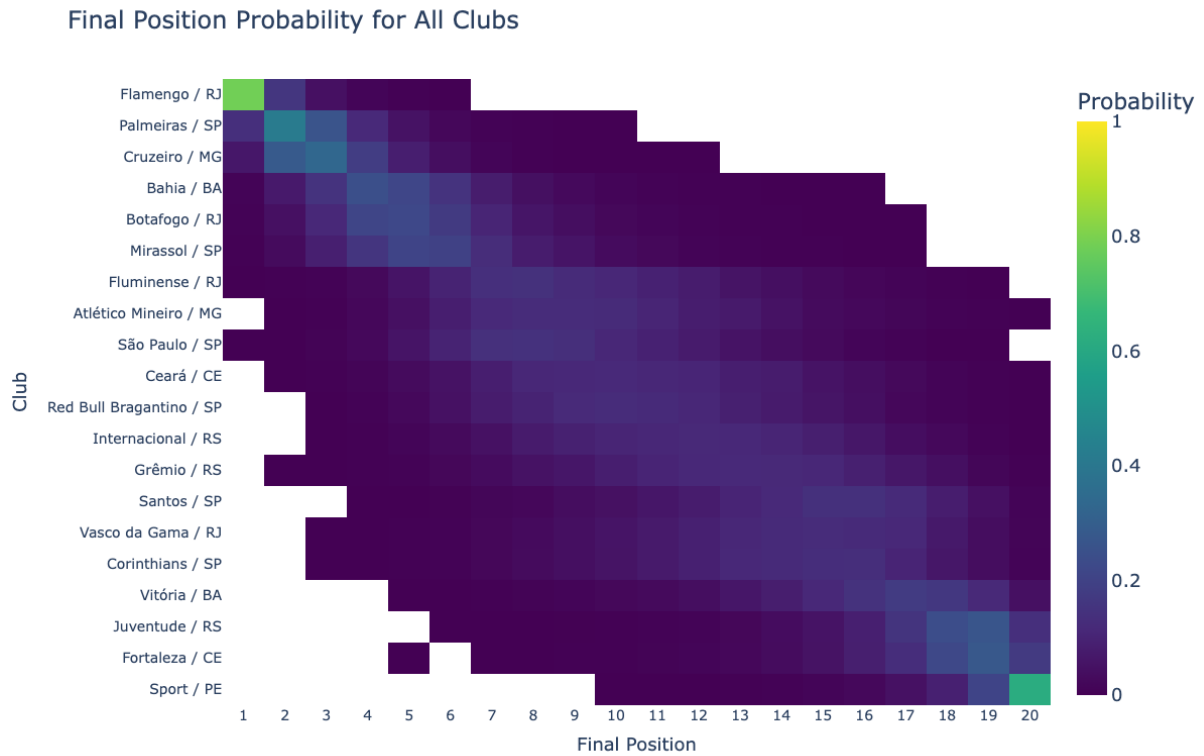


Figure 21 – **Final position probability distribution at mid-season 2025 (after 189 matches)**. Each cell represents the estimated probability of a club (row) finishing the season in a given position (column), based on 10,000 Monte Carlo simulations of the remaining 191 matches. Clubs are ordered vertically by expected final position; brighter colours indicate higher probabilities.

considerable overlap with the two top clubs. Fluminense and Bahia also had a median strength above the average.

In the middle of the table, Atlético Mineiro, Grêmio, São Paulo, Santos, Vasco, Corinthians and Ceará had very similar strength distributions, with a considerable overlap with each other. Red Bull Bragantino, Internacional, Fortaleza and Vitória also had similar distributions, slightly below the mid-table group. At the bottom, Juventude and Sport had the lowest median strengths, with a distribution entirely below the average.

5.5 Player-Level Models

This section extends the P7 framework from the club level to the player level, decomposing team strength into individual contributions. While club-level models treat teams as atomic entities, the player-level specification allocates team performance to the players who participated in each match. In this model, each player is assigned a latent log-skill parameter α_p , constrained under a sum-to-zero identifiability condition. This constraint ensures that estimated abilities represent deviations from the league-wide average player, rather than absolute or context-free measures of talent.

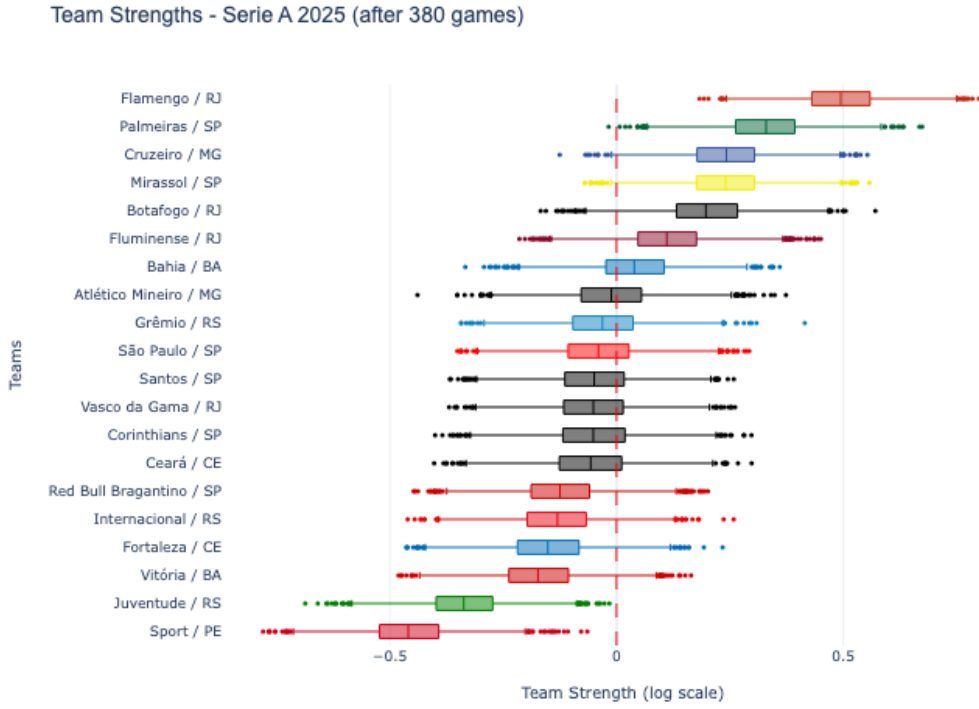


Figure 22 – **Posterior distributions of team strengths for the 2025 *Brasileirão* (after 380 matches)**. Each boxplot shows the posterior distribution of the team strength parameter (log scale) estimated from 4,000 MCMC samples. Teams are ordered vertically by posterior mean strength, from lowest (bottom) to highest (top). The box represents the interquartile range (IQR), the vertical line inside the box indicates the median, and whiskers extend to 1.5 times the IQR. The dashed red vertical line at zero represents the average strength.

To evaluate how temporal aggregation affects estimation, we consider two time periods. The first one corresponds to the 2025 season, the latest full season of the *Brasileirão*. The second one covers the 2020–2025 period, analysed in Section 5.4. For each period, the model estimates individual log-skill parameters for all players who participated in at least one match, along with three global parameters. The global parameters measure the home effect (ν), the common effect parameter (η), and the impact of a red card by the introduction of a synthetic player parameter ($\alpha_{\text{synthetic}}$).

Player rankings are reported using posterior medians of α_p for each player. For descriptive purposes, we also report total minutes played, number of matches, wins, draws, and a set of weighted metrics based on playing time. In these weighted metrics we use the fraction of minutes played by the player in the match to compute their impact on points, goals, and goals difference. For example, a player who played 30 minutes of a match receives 33.3% of the team’s points, goals, and goals difference for that match. Also, to help identify the player, the column “Team” reports the team where the player played the most minutes during the analysed period.

5.5.1 2025 Season

Section 5.4.3 analysed the 2025 *Brasileirão* from a club-level perspective, showing how Flamengo and Palmeiras contested the title throughout the season, with Flamengo securing the championship in the penultimate round. The relegation battle extended to the final match-day, with Ceará and Fortaleza joining Sport and Juventude in the bottom four. Here, we extend this analysis to the player level, decomposing the 20 teams' performance into 730 unique players participating across the 380 matches. The first point of interest is given by the global parameters, in Table 10.

Parameter	Mean	Median	Std	95% CI
ν (home effect)	0.422	0.422	0.066	[0.294, 0.549]
η (common effect)	-0.002	-0.002	0.051	[-0.104, 0.096]
$\alpha_{\text{synthetic}}$	-2.51	-2.28	1.69	$(-\infty, -0.16]$

Table 10 – **Posterior estimates for global parameters under the P7 model for the 2025 season.** Estimates are computed from 30,000 MCMC samples.

The home effect estimate of $\nu = 0.422$ corresponds to a multiplicative factor of $\exp(0.422) \approx 1.52$ on expected goals for the home team. This value is nearly identical to the club-level estimate reported in Table 9 ($\nu = 0.425$), showing consistency between the two levels of analysis. The common effect parameter $\eta \approx 0$ indicates a negligible systematic deviation in overall scoring levels beyond what is explained by team strengths. This value is slightly different from the one estimated at the club-level ($\eta = -0.062$), but both credible intervals include zero.

The new parameter representing the synthetic player, $\alpha_{\text{synthetic}} = -2.28$, quantifies the effect of playing with fewer players due to red cards. Since we are interested in answering whether a red card has a negative impact on club performance, we calculate a unilateral credible interval. Given the credible interval, we can affirm that a red card acts as a below-average player. Compared with other parameters, this estimate has a larger variance because the sum-to-zero constraint defines $\alpha_{\text{synthetic}} = -\sum_p \alpha_p$, causing its uncertainty to accumulate across all individual player estimates.

Beyond global parameters, the model estimates individual skill coefficients for each player. These estimates capture each player's contribution to their team's aggregate strength, as defined by Equation 4.20. This aggregate strength determines the expected goal rate relative to the opponent's strength, after accounting for home effect. Table 11 presents the top 20 players by estimated log-skill.

The ranking is dominated by players from Flamengo (8 of top 20) and Cruzeiro (6), reflecting their strong performances as champion and third-place finisher, respectively. The other players are also from clubs that finished in the top positions of the league, with Mirassol standing out with three players in their first season at *Brasileirão Série A*:

#	Player	Team	Median	95% CI	Min.	G	W	D	WS	WG	WGD
1	Agustin Rossi	Flamengo	0.049	[-0.15, 0.25]	3,330	37	23	9	78.0	75.0	51.0
2	Léo Pereira	Flamengo	0.046	[-0.16, 0.25]	2,954	33	22	8	73.5	68.5	47.5
3	De Arrascaeta	Flamengo	0.037	[-0.16, 0.23]	2,402	33	22	7	58.5	52.2	35.8
4	Léo Ortiz	Flamengo	0.030	[-0.17, 0.23]	2,248	26	16	7	52.0	45.0	31.7
5	Luiz Araújo	Flamengo	0.029	[-0.17, 0.23]	1,712	33	22	7	44.0	40.9	29.3
6	Villalba	Cruzeiro	0.027	[-0.17, 0.23]	2,864	32	16	13	60.5	47.3	27.3
7	Weverton	Palmeiras	0.027	[-0.17, 0.23]	2,340	26	19	3	60.0	49.0	28.0
8	Jemmes	Mirassol	0.026	[-0.17, 0.22]	3,318	37	18	13	66.6	62.7	25.9
9	Lucas Silva	Cruzeiro	0.025	[-0.17, 0.22]	2,752	34	18	12	58.5	45.9	24.6
10	Christian	Cruzeiro	0.025	[-0.17, 0.22]	2,387	30	17	10	55.1	41.2	25.7
11	Kaio Jorge	Cruzeiro	0.025	[-0.18, 0.22]	2,687	33	18	11	58.0	45.3	24.5
12	Fabício Bruno	Cruzeiro	0.024	[-0.17, 0.22]	2,950	33	17	11	61.8	49.0	24.0
13	Kaiki Bruno	Cruzeiro	0.024	[-0.17, 0.22]	2,996	34	17	12	62.3	49.9	24.4
14	Daniel	Mirassol	0.024	[-0.17, 0.22]	3,265	37	18	12	65.5	60.8	23.9
15	Varela	Flamengo	0.024	[-0.18, 0.22]	1,326	20	13	5	36.1	31.6	25.6
16	Pedro	Flamengo	0.023	[-0.17, 0.22]	1,044	21	14	5	27.9	30.6	23.7
17	Vitinho	Botafogo	0.023	[-0.17, 0.22]	2,875	36	17	11	56.7	50.9	21.2
18	Neto	Mirassol	0.023	[-0.18, 0.22]	2,479	34	16	12	50.4	49.0	21.4
19	Alex Sandro	Flamengo	0.023	[-0.17, 0.22]	1,499	19	13	5	37.0	33.2	23.2
20	Joaquin Piquerez	Palmeiras	0.022	[-0.17, 0.22]	2,411	29	18	5	55.9	47.2	24.8

Table 11 – **Top 20 players ranked by estimated skill under the P7 model for the 2025 season.** Skill estimates are posterior medians from 30,000 MCMC samples. Positive values indicate above-average skill relative to the league. Abbreviations: Min. = total minutes played, G = games, W = wins, D = draws, WS = weighted points (sum of team points weighted by player’s share of match minutes), WG = weighted goals, WGD = weighted goal difference.

Jemmes, Daniel and Neto. The top-ranked player, Agustin Rossi, has over 3,300 minutes with 23 wins in 37 matches, helping Flamengo to earn 78 points and the championship. During his participation, Flamengo scored a weighted total of 75 goals with a goal difference of 51.

Just as the top rankings concentrate players from the best performing clubs, the bottom rankings reveal the other extreme, with players from relegated clubs. Some of these players accumulated significant playing time on losing teams, resulting in skill estimates that reflect the team’s poor performance. For example, given Sport’s relegation with only 17 points, their players unsurprisingly dominate the lowest positions, as shown in Table 12.

Beyond Sport, the bottom rankings include some players from Juventude, another club with a poor campaign that was relegated. The lowest-ranked player, Christian Rivera, has over 2,800 minutes across 33 matches, during which Sport won only twice and earned 14 points, with a weighted goal difference of -39.5. Some Sport players recorded zero wins across the entire season: Lucas (11 matches, weighted goal difference of -23.9) and Caique França (15 matches, weighted goal difference of -21.0). In addition to Sport and Juventude players, the bottom 20 also include Ramon and Renato Kayzer, both from Vitória, a club that narrowly avoided relegation.

#	Player	Team	Median	95% CI	Min.	G	W	D	WS	WG	WGD
711	Ewerthon	Juventude	-0.012	[-0.21, 0.18]	1,110	17	3	5	8.0	10.1	-16.7
712	Renato Kayzer	Vitória	-0.013	[-0.21, 0.18]	2,042	28	7	9	21.9	18.4	-18.3
713	Caique	Juventude	-0.014	[-0.21, 0.18]	2,054	28	7	6	18.8	19.5	-20.1
714	Ramon	Vitória	-0.014	[-0.21, 0.18]	1,949	23	7	5	25.1	15.1	-20.1
715	Chrystian	Sport	-0.015	[-0.21, 0.18]	1,546	33	2	11	6.8	10.7	-18.7
716	Ramon Menezes	Sport	-0.015	[-0.21, 0.17]	1,923	23	2	7	13.0	19.5	-22.9
717	Pablo	Sport	-0.015	[-0.21, 0.18]	1,098	20	1	5	2.3	10.1	-21.9
718	Alan Ruschel	Juventude	-0.015	[-0.21, 0.18]	1,275	20	3	4	7.1	8.7	-18.5
719	Caique França	Sport	-0.015	[-0.21, 0.18]	1,350	15	0	4	4.0	8.0	-21.0
720	Igor Carius	Sport	-0.015	[-0.21, 0.18]	1,822	26	2	8	10.2	13.6	-21.5
721	Luiz Gustavo	Juventude	-0.017	[-0.21, 0.18]	2,407	31	6	8	23.9	22.9	-22.6
722	Lucas	Sport	-0.018	[-0.21, 0.17]	904	11	0	0	0.0	6.1	-23.9
723	Luan Candido	Sport	-0.018	[-0.21, 0.18]	1,317	15	1	3	6.0	12.6	-22.6
724	Matheusinho	Sport	-0.019	[-0.21, 0.18]	1,702	23	2	6	10.2	16.3	-23.8
725	Rafael Thyere	Sport	-0.020	[-0.21, 0.18]	2,151	24	2	8	14.0	20.8	-27.8
726	Gabriel Vasconcelos	Sport	-0.021	[-0.21, 0.17]	2,160	24	2	7	13.0	20.0	-28.0
727	Jadson	Juventude	-0.022	[-0.21, 0.17]	2,920	35	9	7	29.9	30.6	-31.4
728	Matheus Alexandre	Sport	-0.023	[-0.22, 0.17]	1,782	25	2	7	9.0	16.1	-28.1
729	Lucas Lima	Sport	-0.028	[-0.22, 0.16]	2,549	32	1	9	10.8	20.6	-36.5
730	Christian Rivera	Sport	-0.028	[-0.22, 0.16]	2,818	33	2	8	14.0	24.3	-39.5

Table 12 – **Bottom 20 players ranked by estimated skill under the P7 model for the 2025 season.** Skill estimates are posterior medians from 30,000 MCMC samples. Abbreviations: Min. = total minutes played, G = games, W = wins, D = draws, WS = weighted points (sum of team points weighted by player’s share of match minutes), WG = weighted goals, WGD = weighted goal difference.

5.5.2 2020–2025 Period

While the single-season analysis provides a snapshot of player performance, extending the temporal window addresses two key limitations: small sample sizes for individual players and the confounding influence of short-term team dynamics. The six-season period (2020–2025) increases both the sample size and the number of estimated player effects, with 1,975 unique players across 2,280 matches. Before examining individual rankings, we first analyse the global parameter estimates. Table 13 presents the posterior summaries for this extended period.

Parameter	Median	Mean	Std	95% CI
ν (home effect)	0.324	0.324	0.027	[0.270, 0.377]
η (common effect)	0.016	0.016	0.021	[-0.025, 0.057]
$\alpha_{\text{synthetic}}$	-4.81	-5.24	2.42	$(-\infty, -2.07]$

Table 13 – **Posterior estimates for global parameters under the P7 model for the 2020–2025 period.** Estimates are computed from 30,000 MCMC samples.

The home effect ($\nu = 0.324$) corresponds to a multiplicative factor of approximately 1.38 on expected goals, consistent with the range of club-level estimates in Table 9 (log between 0.273 and 0.425, factor between 1.31 and 1.53). The common effect parameter η remains indistinguishable from zero, as in the single-season analysis. The synthetic player parameter ($\alpha_{\text{synthetic}} = -4.81$) is considerably lower than the 2025 estimate (-2.28), and its credible interval now excludes zero by a wide margin. In practical terms, this implies

that receiving a red card is equivalent to replacing a player with one whose relative skill is $\exp(-4.81) \approx 0.008$ times the average.

With global parameters consistent with both the single-season and club-level analyses, we turn to individual player estimates. The multi-season aggregation produces more stable skill coefficients, as players accumulate more minutes across multiple campaigns. Tables 14 and 15 present the top and bottom 20 players ranked by posterior median log-skill, respectively.

#	Player	Team	Median	95% CI	Min.	G	W	D	WS	WG	WGD
1	Weverton	Palmeiras	0.132	[-0.07, 0.33]	16,448	183	105	44	359.0	308.0	153.5
2	De Arrascaeta	Flamengo	0.093	[-0.10, 0.30]	9,864	142	87	29	225.9	205.5	106.7
3	Fabício Bruno	Flamengo	0.083	[-0.12, 0.28]	13,391	155	81	43	275.7	235.6	94.2
4	Gustavo Gómez	Palmeiras	0.083	[-0.12, 0.29]	13,786	157	86	39	290.7	236.2	105.5
5	Murilo	Palmeiras	0.082	[-0.12, 0.28]	8,930	102	60	27	202.5	167.3	96.4
6	Joaquín Piquerez	Palmeiras	0.079	[-0.12, 0.28]	9,453	112	68	24	216.5	179.2	99.9
7	Marcos Rocha	Palmeiras	0.076	[-0.12, 0.27]	9,723	133	75	34	213.0	178.8	91.1
8	Raphael Veiga	Palmeiras	0.076	[-0.13, 0.28]	11,203	164	91	37	236.4	200.3	93.0
9	Léo Pereira	Flamengo	0.067	[-0.13, 0.27]	11,332	136	72	35	231.1	207.6	84.0
10	Zé Rafael	Palmeiras	0.064	[-0.14, 0.26]	11,385	159	84	39	232.3	199.9	82.3
11	Agustin Rossi	Flamengo	0.063	[-0.14, 0.26]	7,830	87	49	22	169.0	149.0	74.0
12	Éverson	Atlético-MG	0.060	[-0.14, 0.26]	17,901	199	90	57	326.9	276.8	72.0
13	Renê	Internacional	0.058	[-0.14, 0.26]	9,691	128	67	33	197.1	152.3	61.0
14	Rony	Palmeiras	0.056	[-0.14, 0.25]	11,222	181	90	49	221.0	190.4	73.1
15	Léo Ortiz	RB Bragantino	0.055	[-0.15, 0.25]	12,565	149	65	48	226.6	202.7	64.7
16	Pedro	Flamengo	0.052	[-0.15, 0.25]	9,030	157	87	33	184.7	170.2	65.9
17	Bruno Henrique	Flamengo	0.052	[-0.15, 0.25]	10,385	148	81	35	211.8	182.4	64.7
18	Ayrton Lucas	Flamengo	0.052	[-0.15, 0.25]	8,812	116	60	26	174.8	159.9	61.6
19	Gerson	Flamengo	0.050	[-0.15, 0.25]	8,946	109	61	25	192.2	173.4	63.5
20	Victor Cuesta	Internacional	0.050	[-0.15, 0.25]	12,449	141	63	40	226.6	192.8	55.3

Table 14 – **Top 20 players ranked by estimated skill under the P7 model for the 2020–2025 period.** Skill estimates are posterior medians from 30,000 MCMC samples. Abbreviations: Min. = total minutes played, G = games, W = wins, D = draws, WS = weighted points (sum of team points weighted by player’s share of match minutes), WG = weighted goals, WGD = weighted goal difference.

Two points immediately stand out from the rankings. First, all credible intervals (both top and bottom and in all periods) include zero, meaning the model cannot statistically distinguish any individual player from the league-wide average. Second, the top 20 is heavily concentrated in two clubs: Palmeiras and Flamengo, with 8 players each.

Despite the statistical indistinguishability, examining the ranking by posterior median reveals interpretable patterns. In this period, Palmeiras was champion twice, in 2022 and 2023, and finished in the top half of the table in the other seasons: 7th in 2020, 3rd in 2021 and 2nd in both 2024 and 2025. Flamengo also won the championship twice, in 2020 and 2025, with good results in the other seasons: 2nd in 2021, 5th in 2022, 4th in 2023 and 3rd in 2024.

The four remaining players in the top 20 (Éverson, Renê, Léo Ortiz, and Victor Cuesta) also accumulated significant minutes in successful environments, though across multiple clubs. Éverson won the 2021 title with Atlético-MG and finished in the top

#	Player	Team	Median	95% CI	Min.	G	W	D	WS	WG	WGD
1,956	Victor Luis	Botafogo	-0.026	[-0.22, 0.17]	6,632	104	23	31	67.1	74.1	-32.7
1,957	Lucas	América-MG	-0.026	[-0.22, 0.17]	7,119	94	27	20	89.6	78.8	-34.7
1,958	Rodrigo Alves	Juventude	-0.026	[-0.22, 0.16]	2,788	35	3	11	17.9	23.0	-34.4
1,959	Pedro Raul	Ceará	-0.026	[-0.22, 0.16]	8,898	123	27	39	101.1	93.5	-37.9
1,960	Victor Hugo	Bahia	-0.027	[-0.22, 0.16]	10,710	124	37	27	132.6	128.9	-36.8
1,961	Shaylon	Atlético-GO	-0.028	[-0.22, 0.17]	6,143	103	26	28	65.1	63.3	-36.6
1,962	Rafael Thyere	Sport	-0.028	[-0.22, 0.17]	5,567	66	14	20	56.0	49.1	-37.2
1,963	Lucas Halter	Goiás	-0.029	[-0.22, 0.16]	8,664	104	33	27	112.1	94.5	-39.1
1,964	Christian Rivera	Sport	-0.029	[-0.22, 0.16]	2,818	33	2	8	14.0	24.3	-39.5
1,965	Gilberto	Bahia	-0.030	[-0.22, 0.16]	8,156	133	37	30	93.8	99.0	-38.8
1,966	Matheus	Cuiabá	-0.030	[-0.22, 0.16]	8,629	111	27	26	96.8	91.8	-37.9
1,967	Gabriel Baralhas	Atlético-GO	-0.031	[-0.23, 0.16]	9,088	129	37	35	110.1	96.7	-44.6
1,968	Éder	Atlético-GO	-0.034	[-0.22, 0.16]	10,617	126	37	35	136.3	111.3	-46.0
1,969	Tadeu	Goiás	-0.035	[-0.22, 0.15]	9,450	105	26	31	109.0	107.0	-48.0
1,970	Matheusinho	Sport	-0.036	[-0.23, 0.16]	4,960	85	20	20	52.5	55.4	-41.3
1,971	Isidro Pitta	Cuiabá	-0.037	[-0.23, 0.15]	7,161	118	31	27	66.6	64.7	-48.8
1,972	Kevin	Avaí	-0.037	[-0.23, 0.15]	5,213	65	11	15	45.4	51.6	-47.6
1,973	Lucas Lima	Sport	-0.044	[-0.24, 0.15]	8,823	137	32	40	91.4	93.5	-54.0
1,974	Jadson	Juventude	-0.050	[-0.24, 0.14]	10,300	131	30	38	113.0	114.7	-66.4
1,975	Gabriel Vasconcelos	Coritiba	-0.051	[-0.24, 0.14]	9,110	102	23	25	93.5	104.2	-71.4

Table 15 – **Bottom 20 players ranked by estimated skill under the P7 model for the 2020–2025 period.** Skill estimates are posterior medians from 30,000 MCMC samples. Abbreviations: Min. = total minutes played, G = games, W = wins, D = draws, WS = weighted points (sum of team points weighted by player’s share of match minutes), WG = weighted goals, WGD = weighted goal difference.

three twice more. Renê played for Flamengo (2020–2021), Internacional (2022–2024), and Fluminense (2025), consistently participating in top-half finishes. Léo Ortiz spent four seasons at RB Bragantino before transferring to Flamengo in 2024, while Victor Cuesta played for Internacional, Botafogo, and Bahia, all clubs that achieved mid-to-upper table finishes during his tenure. Conversely, the bottom 20 includes players from a broader set of teams that experienced relegation or persistent struggles: Coritiba, Juventude, Sport, Cuiabá, Avaí, Goiás, and Atlético Goianiense. These patterns indicate that posterior rankings reflect sustained exposure to winning or losing contexts rather than isolated individual excellence or deficiency.

This interpretation is reinforced by the near-perfect correlation ($r = 0.99$) between posterior medians and weighted goal difference (WGD). Since WGD aggregates team-level outcomes (goals scored minus goals conceded) weighted by each player’s share of match minutes, this correlation reveals that estimated skill largely captures whether a player recorded minutes in winning or losing environments. In the end, the model effectively transforms collective team performance into individual-level estimates, with limited capacity to isolate marginal contributions.

This structural dependence has notable consequences for certain player profiles. Goalkeepers, who typically play full matches whenever fielded, are particularly susceptible to extreme rankings because their WGD closely tracks their team’s aggregate performance. Among the top 20, three goalkeepers appear: Weverton (1st), Agustin Rossi (11th), and Éverson (12th). Among the bottom 20, Tadeu (1,969th) and Gabriel Vasconcelos (1,975th)

occupy similarly extreme positions. The pattern extends beyond goalkeepers to any player who consistently appears in the starting lineup of a club and plays a considerable fraction of the match.

The case of Pedro Raul illustrates this dynamic. Despite being the runner-up top scorer of the 2022 *Brasileirão* while playing for Goiás (13th place in that season) and scoring 11 goals for Ceará in 2025, he ranks 1,959th out of 1,975 players in the six-season analysis. Across 123 matches and nearly 9,000 minutes, he participated in only 27 wins and recorded a weighted goal difference of -37.9 . These numbers reflect the poor performance of teams like Goiás and Ceará rather than his individual ability. Since the model observes only match outcomes, his individual attacking output cannot offset the collective struggles of his teams.

These findings echo the club-level analysis of Section 5.4. Just as team strength estimates reflect aggregate match outcomes, player-level estimates reflect the teams in which players played. The decomposition from clubs to players increases model dimensionality but does not resolve the fundamental challenge of separating individual ability from collective performance. Future work incorporating player-specific actions or event data may offer a path toward more granular attribution.

6 Conclusion

Probabilistic models for football outcomes have been studied extensively, specifically for European competitions. However, the validation via Simulation-Based Calibration and the extension to the player level remain uncommon. This work addresses this gap by making three main contributions.

The first contribution is a dataset spanning from 2013 to 2025, covering five competitions (*Série A* through *D* and *Copa do Brasil*). It maps over 16,800 unique players across 368 clubs and is available at <https://github.com/BrazilianFootball/Data>, with an archived version on Zenodo (MICHELS; CARVALHO, 2026). The extraction pipeline converts PDF match reports into structured JSON files with match-level summary and temporal segmentation based on substitutions and red cards. This information enables the analysis at both club and player levels.

The second contribution is the application of Simulation-Based Calibration to validate the inference procedure for all 22 proposed models. The validation occurs for 14 club-level (4 Bradley-Terry and 10 Poisson variants) and 8 player-level (all Poisson variants) specifications. All of them passed validation, with flagging rates consistent with chance under 95% confidence bands. This procedure ensures that posterior distributions are correctly calibrated before application to real data.

The third contribution is the comparative evaluation of models on *Brasileirão* data from 2020–2025 and their extension to the player level. Model P7 (single skill with home effect and common effect) achieved the best predictive performance according to Brier Score, Log Score, and Ranked Probability Score. Analysing year by year, the home effect remained robust over time, with estimates for the multiplicative effect on the home team’s expected goals ranging from 1.31 to 1.53. The common effect parameter was statistically indistinguishable from zero, indicating that team strengths alone adequately capture the goal-scoring structure in *Brasileirão*.

At the player level, log-skill parameters were estimated for 730 unique players in the single-season analysis and 1,975 in the multi-season analysis. The synthetic player parameter quantifying red card effects showed strong negative impact on the team’s expected goals when aggregating players to the team level. For regular players, all estimated credible intervals include zero, indicating that the model fails to identify any significant difference between players. Furthermore, individual skill estimates correlate near-perfectly with weighted goal difference. This is a strong indication that the player-level model captures the team context in matches played by the player rather than isolated individual ability.

This confounding between player skill and team performance reflects a fundamental

limitation of models relying solely on match outcomes. Without individual action data, the contribution of each player cannot be separated from the collective team effect. More broadly, the models assume that team strengths are constant within each season and that matches are exchangeable. This assumption ignores several factors that may influence outcomes, such as fatigue from fixture congestion and the differential pressure of decisive versus inconsequential matches. Under exchangeability, a win in the first round carries the same inferential weight as a win in the final round, precluding the detection of form fluctuations or momentum effects.

Several directions could address these limitations. Incorporating player-specific event data would enable finer performance attribution and reduce the confounding between individual skill and team context. Relaxing the exchangeability assumption would require dynamic specifications where team and player abilities evolve over time. State-space formulations, in which parameters follow a random walk or autoregressive process, would allow the model to capture form fluctuations. Such extensions would also enable predictions that account for recent performance rather than treating all historical matches equally. Another possibility is adding contextual covariates, such as days since last match, travel distance, or an indicator for decisive matches, to the model. These covariates would allow the model to distinguish between high- and low-pressure moments without abandoning the exchangeability framework entirely.

To support reproducibility, all code used in this work is available at <https://github.com/BrazilianFootball/Modeling>.

References

ALDOUS, David. Elo Ratings and the Sports Model: A Neglected Topic in Applied Probability? **Statistical Science**, Institute of Mathematical Statistics, v. 32, n. 4, p. 616–629, 2017. DOI: [10.1214/17-STS628](https://doi.org/10.1214/17-STS628).

ARRUDA, Marcelo Leme de. **Chance de Gol**. 1999. Available from: [<https://www.chancedegol.com.br/>](https://www.chancedegol.com.br/).

BAIO, Gianluca; BLANGIARDO, Marta. Bayesian hierarchical model for the prediction of football results. **Journal of Applied Statistics**, v. 37, n. 2, p. 253–264, 2010.

BENZ, Luke S.; LOPEZ, Michael J. Estimating the change in soccer's home advantage during the Covid-19 pandemic using bivariate Poisson regression. **AStA Advances in Statistical Analysis**, v. 107, n. 1, p. 205–232, 2023. DOI: [10.1007/s10182-021-00413-9](https://doi.org/10.1007/s10182-021-00413-9).

BRADLEY, Ralph Allan; TERRY, Milton E. Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons. **Biometrika**, [Oxford University Press, Biometrika Trust], v. 39, n. 3/4, p. 324–345, 1952. ISSN 00063444, 14643510. Available from: [<http://www.jstor.org/stable/2334029>](http://www.jstor.org/stable/2334029).

CARPENTER, Bob et al. Stan: A probabilistic programming language. **Journal of Statistical Software**, Columbia Univ., New York, NY (United States); Harvard Univ., Cambridge, MA (United States), v. 76, n. 1, 2017.

CATTELAN, Manuela; VARIN, Cristiano; FIRTH, David. Dynamic Bradley–Terry modelling of sports tournaments. **Journal of the Royal Statistical Society: Series C (Applied Statistics)**, v. 62, n. 1, p. 135–150, 2013. DOI:

[10.1111/j.1467-9876.2012.01046.x](https://doi.org/10.1111/j.1467-9876.2012.01046.x). Available from: [<https://rss.onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9876.2012.01046.x>](https://rss.onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9876.2012.01046.x).

COOK, Samantha R.; GELMAN, Andrew; RUBIN, Donald B. Validation of Software for Bayesian Models Using Posterior Quantiles. **Journal of Computational and Graphical Statistics**, [American Statistical Association, Taylor and Francis, Ltd., Institute of Mathematical Statistics, Interface Foundation of America], v. 15, n. 3, p. 675–692, 2006. ISSN 10618600. Available from: [<http://www.jstor.org/stable/27594203>](http://www.jstor.org/stable/27594203).

DAVIDSON, Roger R. On Extending the Bradley-Terry Model to Accommodate Ties in Paired Comparison Experiments. **Journal of the American Statistical Association**, [American Statistical Association, Taylor and Francis, Ltd.], v. 65, n. 329, p. 317–328, 1970. ISSN 01621459, 1537274X. Available from: [<http://www.jstor.org/stable/2283595>](http://www.jstor.org/stable/2283595).

- DAVIDSON, Roger R.; BEAVER, Robert J. On Extending the Bradley-Terry Model to Incorporate Within-Pair Order Effects. **Biometrics**, International Biometric Society, v. 33, n. 4, p. 693–702, 1977. ISSN 0006341X, 15410420. Available from: <http://www.jstor.org/stable/2529467>.
- DEPARTAMENTO DE MATEMÁTICA, UFMG. **Probabilidades no Futebol**. 2005. Available from: <https://www.mat.ufmg.br/futebol/>.
- DINIZ, Marcio Alves et al. Comparing probabilistic predictive models applied to football. **Journal of the Operational Research Society**, Informa UK Limited, v. 70, n. 5, p. 770–782, Apr. 2018. ISSN 1476-9360. DOI: [10.1080/01605682.2018.1457485](https://doi.org/10.1080/01605682.2018.1457485). Available from: <http://dx.doi.org/10.1080/01605682.2018.1457485>.
- DIXON, Mark J.; COLES, Stuart G. Modelling Association Football Scores and Inefficiencies in the Football Betting Market. **Journal of the Royal Statistical Society. Series C (Applied Statistics)**, [Royal Statistical Society, Oxford University Press], v. 46, n. 2, p. 265–280, 1997. ISSN 00359254, 14679876. Available from: <http://www.jstor.org/stable/2986290>.
- ELO, Arpad E. The USCF Rating System – A Scientific Achievement. **Chess Life**, United States Chess Federation, v. 16, n. 6, p. 160–161, June 1961.
- FISCHER, Kai; HAUCAP, Justus. Does Crowd Support Drive the Home Advantage in Professional Football? Evidence from German Ghost Games during the COVID-19 Pandemic. **Journal of Sports Economics**, v. 22, n. 8, 2021. DOI: [10.1177/15270025211026552](https://doi.org/10.1177/15270025211026552).
- GELMAN, Andrew et al. **Bayesian Workflow**. [S.l.: s.n.], 2020. arXiv: [2011.01808](https://arxiv.org/abs/2011.01808) [stat.ME]. Available from: <https://arxiv.org/abs/2011.01808>.
- GRAND VIEW RESEARCH. **Sports Betting Market Size & Share, Industry Report, 2030**. 2024. Available from: <https://www.grandviewresearch.com/industry-analysis/sports-betting-market-report>. Visited on: 22 Feb. 2026.
- HOLGATE, P. Estimation for the bivariate Poisson distribution. **Biometrika**, v. 51, n. 1-2, p. 241–287, June 1964. ISSN 0006-3444. DOI: [10.1093/biomet/51.1-2.241](https://doi.org/10.1093/biomet/51.1-2.241). Available from: <https://doi.org/10.1093/biomet/51.1-2.241>.
- KARLIS, Dimitris; NTZOUFRAS, Ioannis. Analysis of Sports Data by Using Bivariate Poisson Models. **Journal of the Royal Statistical Society. Series D (The Statistician)**, [Royal Statistical Society, Wiley], v. 52, n. 3, p. 381–393, 2003. ISSN 00390526, 14679884. Available from: <http://www.jstor.org/stable/4128211>.
- _____. Bayesian modelling of football outcomes: using the Skellam’s distribution for the goal difference. **IMA Journal of Management Mathematics**, v. 20, n. 2, p. 133–145, 2009. DOI: [10.1093/imaman/dpn026](https://doi.org/10.1093/imaman/dpn026).

- KARLIS, Dimitris; NTZOUFRAS, Ioannis. On Modelling Soccer Data. **Student**, 2000.
- KOOPMAN, Siem Jan; LIT, Rutger. A dynamic bivariate Poisson model for analysing and forecasting match results in the English Premier League. **Journal of the Royal Statistical Society: Series A**, v. 178, n. 1, p. 167–186, 2015.
- LEE, Alan J. Modeling Scores in the Premier League: Is Manchester United Really the Best? **CHANCE**, ASA Website, v. 10, n. 1, p. 15–19, Jan. 1997. ISSN 0933-2480. DOI: [10.1080/09332480.1997.10554791](https://doi.org/10.1080/09332480.1997.10554791). Available from: <https://doi.org/10.1080/09332480.1997.10554791>.
- MAHER, M. J. Modelling association football scores. **Statistica Neerlandica**, v. 36, n. 3, p. 109–118, 1982. DOI: [10.1111/j.1467-9574.1982.tb00782.x](https://doi.org/10.1111/j.1467-9574.1982.tb00782.x). Available from: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9574.1982.tb00782.x>.
- MICHELS, Igor Patrício; CARVALHO, Luiz Max. **Brazilian Football Match Report Dataset (2013–2025)**. [S.l.]: Zenodo, 2026. Version 2.0.0. DOI: [10.5281/zenodo.19273806](https://doi.org/10.5281/zenodo.19273806). Available from: <https://zenodo.org/records/19273806>.
- MODRÁK, Martin et al. Simulation-Based Calibration Checking for Bayesian Computation: The Choice of Test Quantities Shapes Sensitivity. **Bayesian Analysis**, Institute of Mathematical Statistics, v. 20, n. 2, June 2025. ISSN 1936-0975. DOI: [10.1214/23-ba1404](https://doi.org/10.1214/23-ba1404). Available from: <http://dx.doi.org/10.1214/23-BA1404>.
- NGUYEN, Quang. Poisson Modeling and Predicting English Premier League Goal Scoring. **The New England Journal of Statistics in Data Science**, New England Statistical Society, p. 95–101, Dec. 2021. ISSN 2693-7166. DOI: [10.51387/21-nejsds1](https://doi.org/10.51387/21-nejsds1). Available from: <http://dx.doi.org/10.51387/21-NEJSDS1>.
- PENN, Mark J.; DONNELLY, Christl A. Analysis of a double Poisson model for predicting football results in Euro 2020. **PLOS ONE**, Public Library of Science, v. 17, n. 5, e0268511, 2022. DOI: [10.1371/journal.pone.0268511](https://doi.org/10.1371/journal.pone.0268511).
- POLLARD, Richard. Home advantage in soccer: A retrospective analysis. **Journal of Sports Sciences**, v. 4, n. 3, p. 237–248, 1986. DOI: [10.1080/02640418608732122](https://doi.org/10.1080/02640418608732122).
- PYPDF2 DEVELOPMENT TEAM. **PyPDF2: A Pure-Python PDF Library**. [S.l.: s.n.], 2024. Version 3.0.1. Available from: <https://pypdf2.readthedocs.io/>.
- PYTHON SOFTWARE FOUNDATION. **re — Regular expression operations**. [S.l.], 2024. Python Standard Library. Available from: <https://docs.python.org/3/library/re.html>.
- REITZ, KENNETH AND CONTRIBUTORS. **Requests: HTTP for Humans**. [S.l.: s.n.], 2024. Python HTTP Library. Available from: <https://docs.python-requests.org/>.

RUE, Håvard; SALVESEN, Øyvind. Prediction and Retrospective Analysis of Soccer Matches in a League. **Journal of the Royal Statistical Society. Series D (The Statistician)**, [Royal Statistical Society, Wiley], v. 49, n. 3, p. 399–418, 2000. ISSN 00390526, 14679884. Available from: <<http://www.jstor.org/stable/2681065>>.

SORS, Fabrizio et al. The sound of silence in association football: Home advantage and referee bias decrease in matches played without spectators. **European Journal of Sport Science**, v. 21, n. 12, p. 1597–1605, 2021. DOI: [10.1080/17461391.2020.1845814](https://doi.org/10.1080/17461391.2020.1845814).

STAN DEVELOPMENT TEAM. **CmdStanPy: Python interface to CmdStan**. [S.l.: s.n.], 2023. Version 1.2.0. Available from: <<https://github.com/stan-dev/cmdstanpy>>.

TALTS, Sean et al. **Validating Bayesian Inference Algorithms with Simulation-Based Calibration**. [S.l.: s.n.], 2020. arXiv: [1804.06788](https://arxiv.org/abs/1804.06788) [stat.ME]. Available from: <<https://arxiv.org/abs/1804.06788>>.

WOLF, Stephan; SCHMITT, Maximilian; SCHULLER, Björn. A football player rating system. **Journal of Sports Analytics**, v. 6, p. 1–15, Oct. 2020. DOI: [10.3233/JSA-200411](https://doi.org/10.3233/JSA-200411).

Appendix

APPENDIX A – Additional Season-Level Simulations

This appendix presents simulation graphics for seasons not detailed in the main text. For each season, we show the championship probability evolution, relegation probability evolution, mid-season position probability heatmap, and end-of-season team strength estimates. See Section 5.4 for detailed interpretation of these figures.

A.1 2021 Season

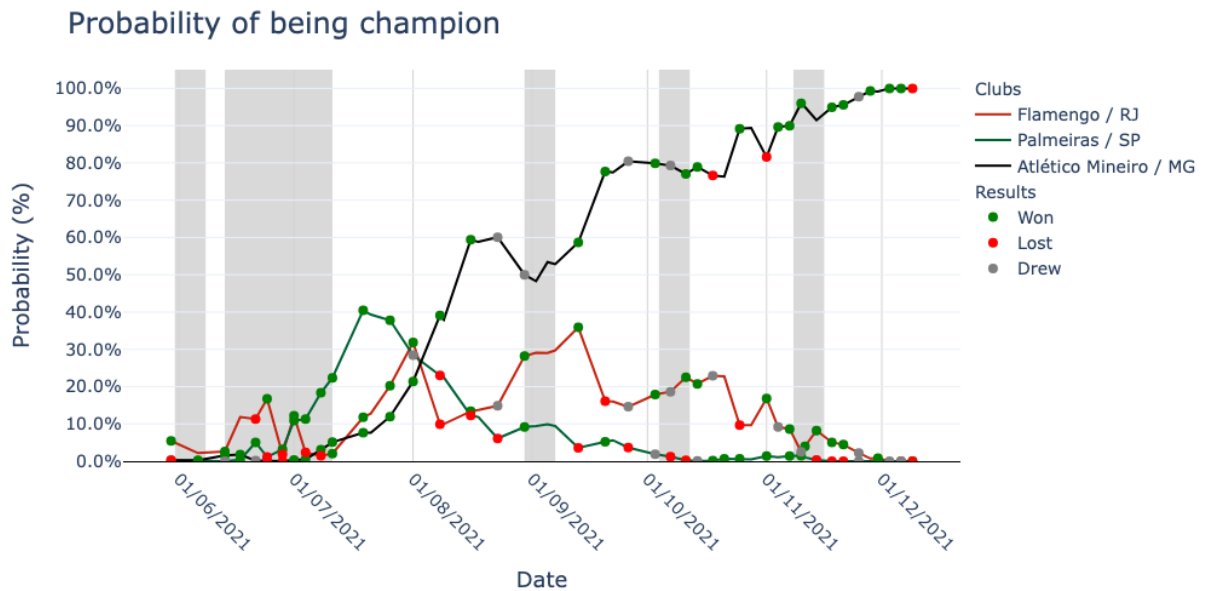


Figure 23 – Championship probability evolution for the 2021 *Brasileirão*.

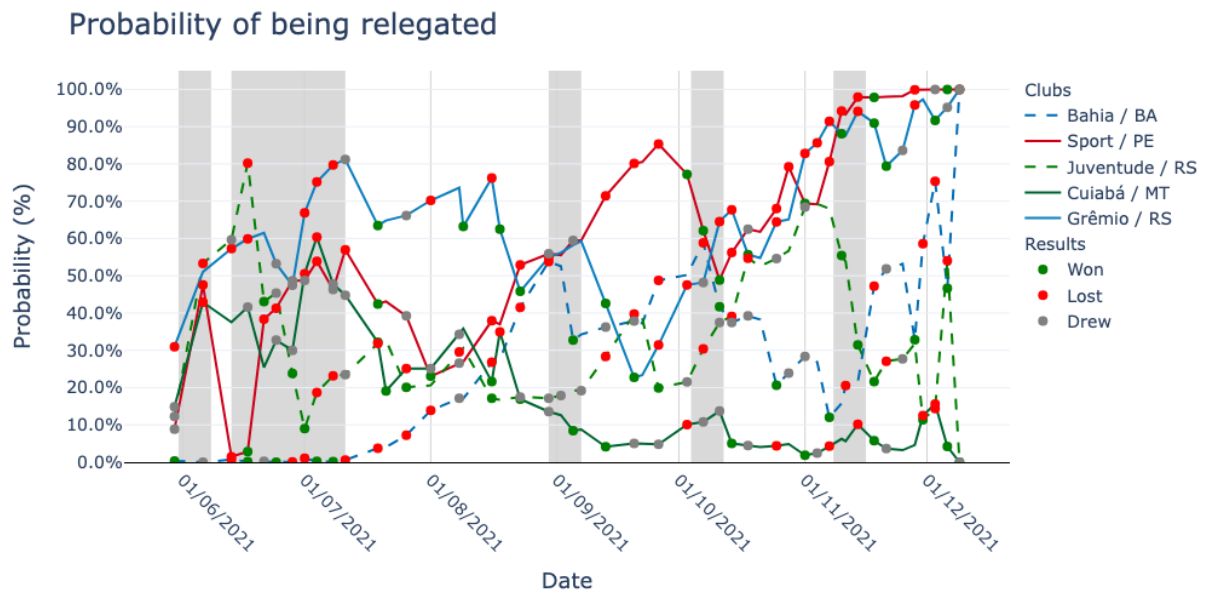
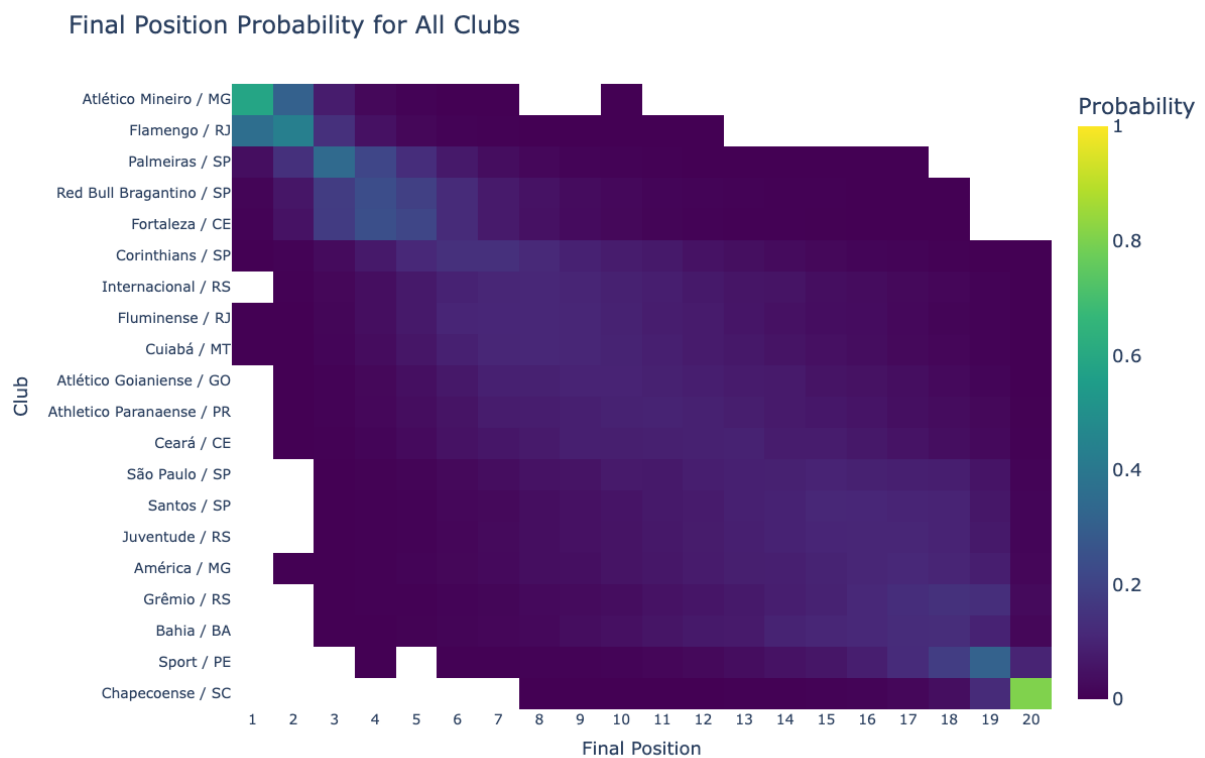
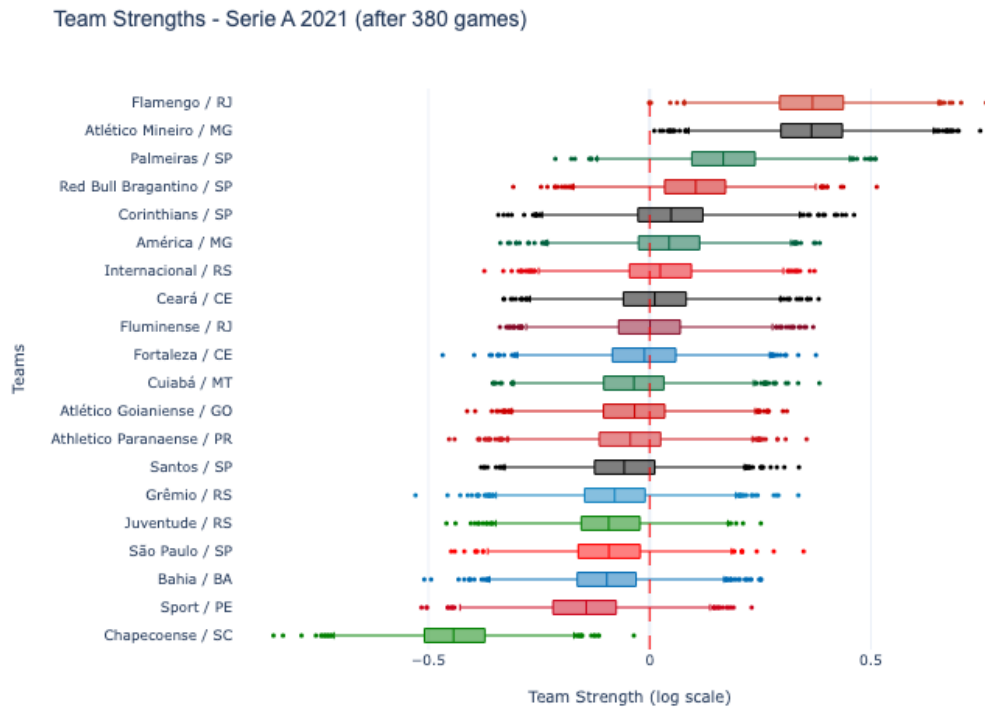
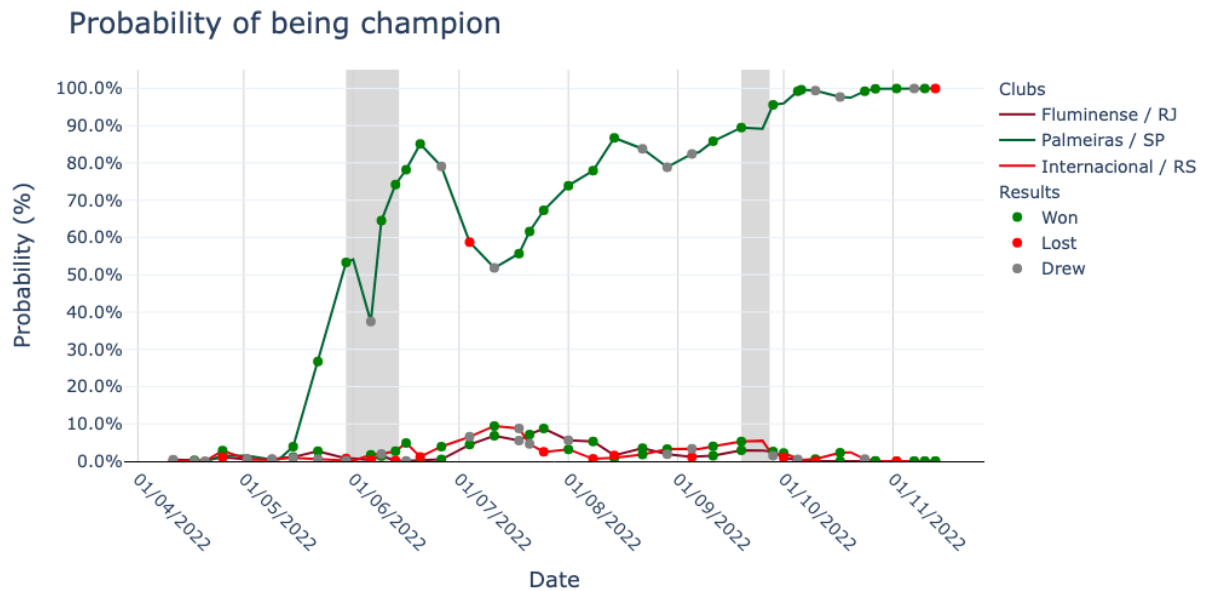
Figure 24 – Relegation probability evolution for the 2021 *Brasileirão*.

Figure 25 – Final position probability distribution at mid-season 2021 (after 193 matches).

Figure 26 – Posterior distributions of team strengths for the 2021 *Brasileirão*.

A.2 2022 Season

Figure 27 – Championship probability evolution for the 2022 *Brasileirão*.

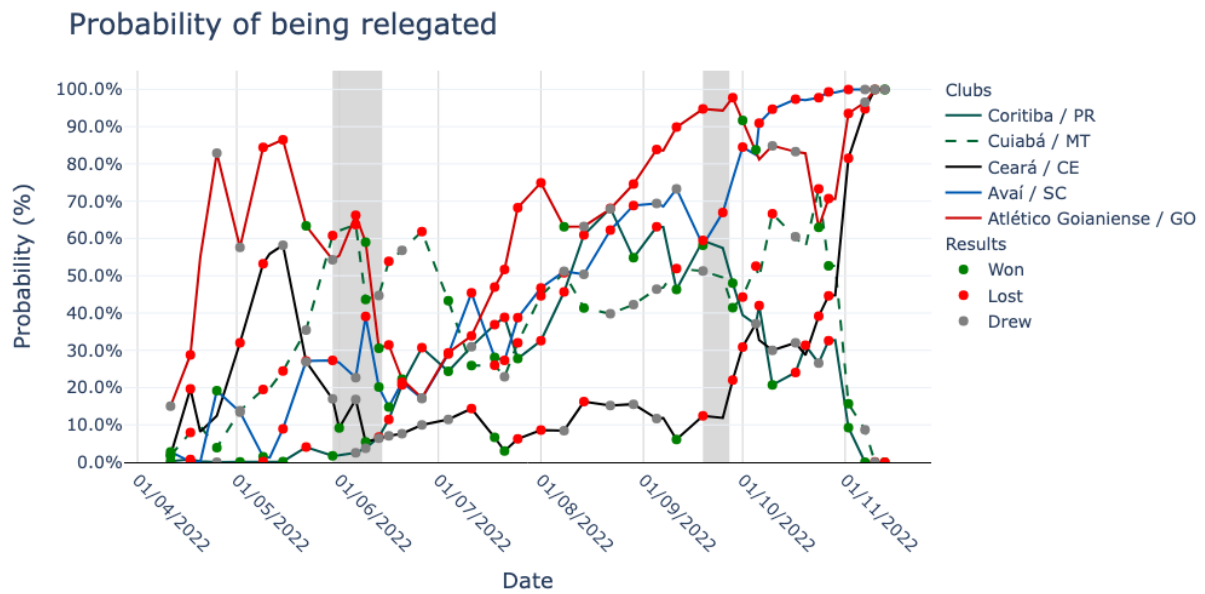
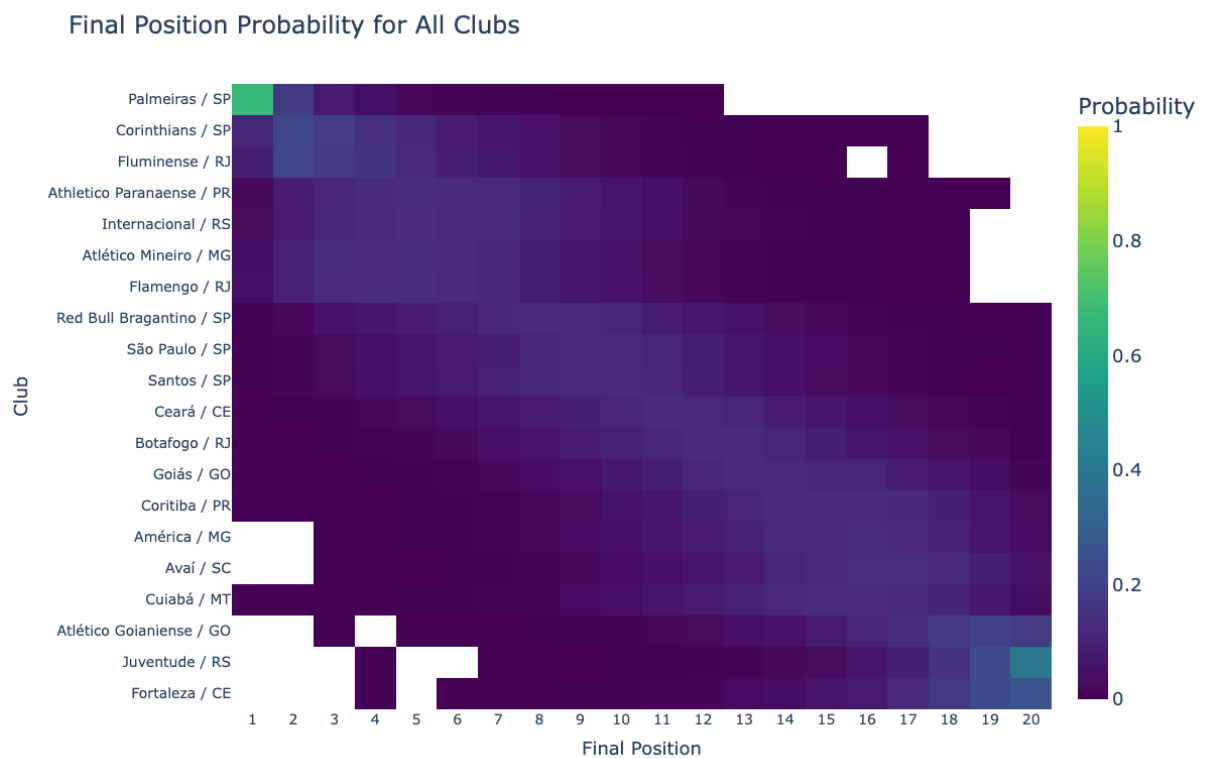
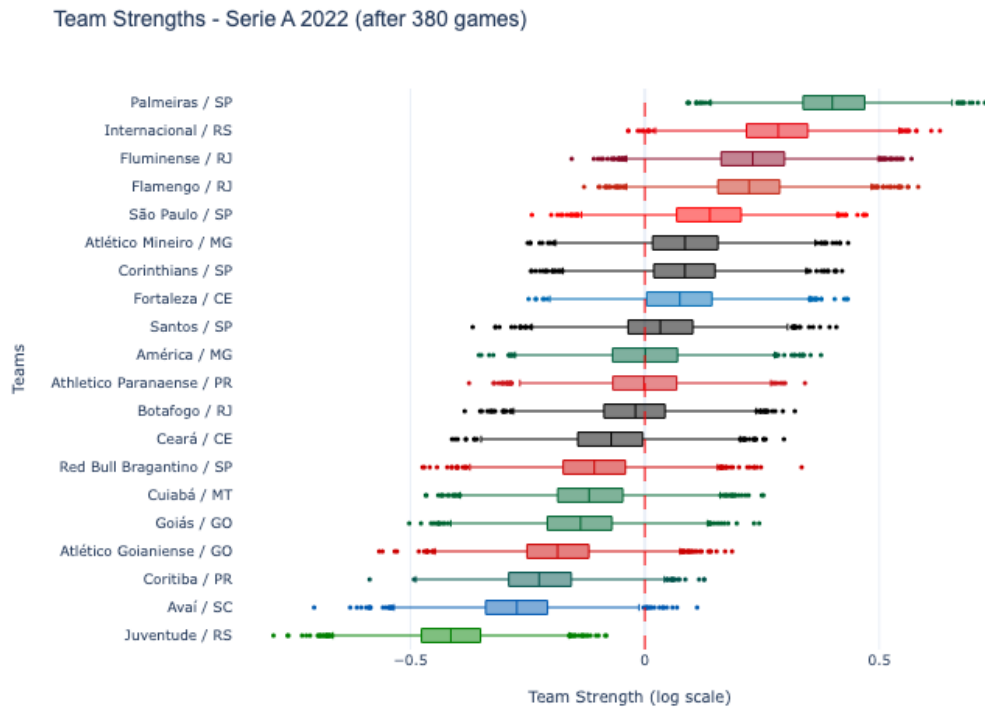
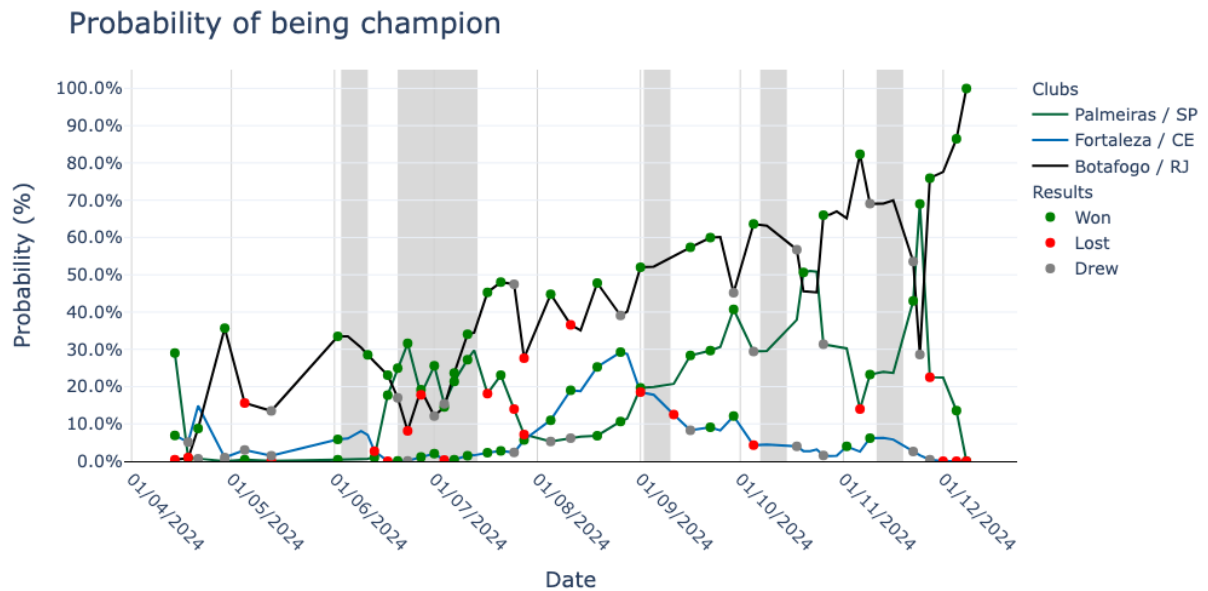
Figure 28 – Relegation probability evolution for the 2022 *Brasileirão*.

Figure 29 – Final position probability distribution at mid-season 2022 (after 190 matches).

Figure 30 – Posterior distributions of team strengths for the 2022 *Brasileirão*.

A.3 2024 Season

Figure 31 – Championship probability evolution for the 2024 *Brasileirão*.

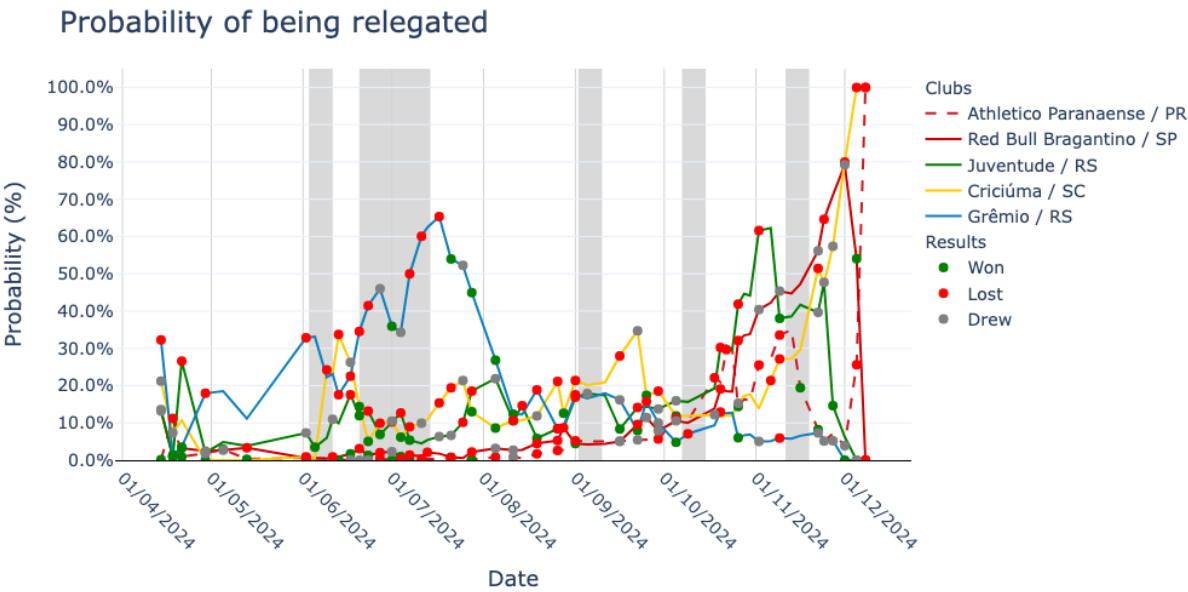


Figure 32 – Relegation probability evolution for the 2024 *Brasileirão*.

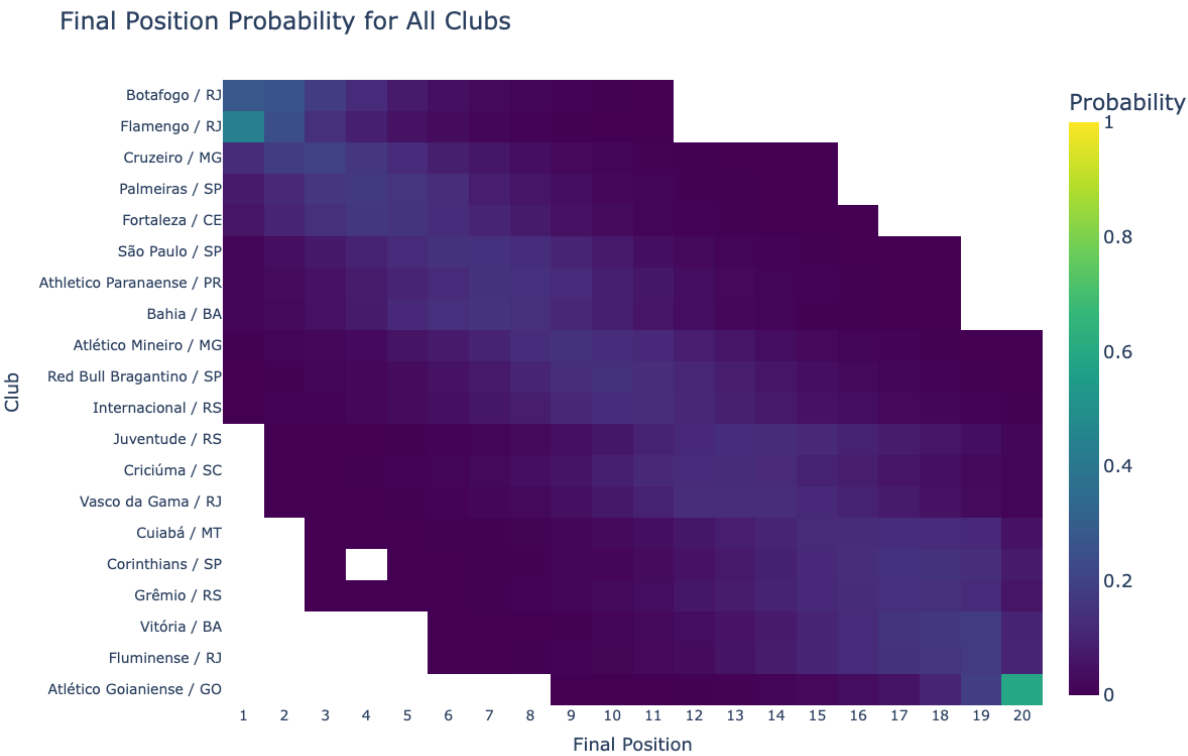
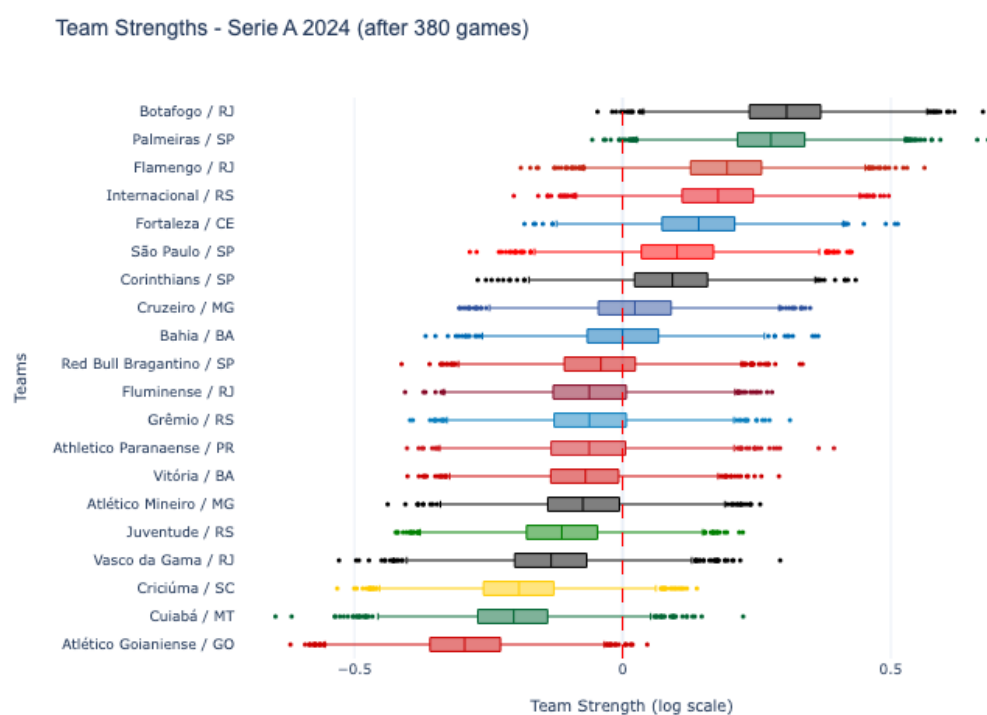


Figure 33 – Final position probability distribution at mid-season 2024 (after 188 matches).

Figure 34 – Posterior distributions of team strengths for the 2024 *Brasileirão*.